



National Taiwan Normal University

# NTNU 2beatdown

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- 1 Contest
- 2 Data structures
- 3 Linear Algebra
- 4 Polynomial
- 5 Number theory
- 6 Graph
- 7 Geometry
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- 10 Mathematics

Contest (1)

template.cpp21 lines

```
#include <bits/stdc++.h>
using namespace std;

#define rep(i, a, b) for(int i=a; i<(b); ++i)
#define all(x) begin(x), end(x)
#define sz(x) (int)(x).size()
#define fi first
#define se second
#define pb push_back

template<typename T> using v = vector<T>;
template<typename T> using vv = v<v<T>>;

typedef long long ll;
typedef pair<int,int> pii;
typedef v<int> vi;

signed main() {
    cin.tie(0)->sync_with_stdio(0);
    cin.exceptions(cin.failbit);
}
```

random.h10c096, 4 lines

```
ll rnd(ll l, ll r) { //[l, r)
    static mt19937_64 gen(chrono::steady_clock::now().
        time_since_epoch().count());
    return uniform_int_distribution<ll>(l, r - 1)(gen);
}
```

一些準則：

1 參考計分版，live scoreboard 8 隊以上開的題目我們理論上也能開，如果實作太困難或覺得怪怪的就求助另一個人。

2 題意理解：確定讀過題目的每個字，確認 sample 也符合你的理解。（在找水題時可以不用 100% 題意理解）。

3 題目交接，如果沒有確保 100% 的題目理解，哪怕有 95%，不要跟另一個人講題目。否則可能賺三分鐘賠兩小時。

4 上機前，確認 100% 的題目理解，並把解題邏輯重新順過，確認沒有漏想的狀況（可以的話把 sample 帶進你的 solution 確認正確性）。這點很難但我們要盡可能減少寫了錯的 solution 情況。

5 抄模板抄完要檢查 1-2 次，可以的話讓另一個人檢查更好。

簡單題有容錯空間，越難的題目越要謹慎遵守，很可能直接影響我們最終題數

troubleshoot.txt52 lines

```
Pre-submit:
Write a few simple test cases if sample is not enough.
Are time limits close? If so, generate max cases.
Is the memory usage fine?
Could anything overflow?
Make sure to submit the right file.

Wrong answer:
Print your solution! Print debug output, as well.
Are you clearing all data structures between test cases?
Can your algorithm handle the whole range of input?
Read the full problem statement again.
Do you handle all corner cases correctly?
Have you understood the problem correctly?
Any uninitialized variables?
Any overflows?
Confusing N and M, i and j, etc.?
Are you sure your algorithm works?
What special cases have you not thought of?
Are you sure the STL functions you use work as you think?
Add some assertions, maybe resubmit.
Create some testcases to run your algorithm on.
Go through the algorithm for a simple case.
Go through this list again.
Explain your algorithm to a teammate.
Ask the teammate to look at your code.
Go for a small walk, e.g. to the toilet.
Is your output format correct? (including whitespace)
Rewrite your solution from the start or let a teammate do it.

Runtime error:
Have you tested all corner cases locally?
Any uninitialized variables?
Are you reading or writing outside the range of any vector?
Any assertions that might fail?
Any possible division by 0? (mod 0 for example)
Any possible infinite recursion?
Invalidated pointers or iterators?
Are you using too much memory?
Debug with resubmits (e.g. remapped signals, see Various).

Time limit exceeded:
Do you have any possible infinite loops?
What is the complexity of your algorithm?
Are you copying a lot of unnecessary data? (References)
How big is the input and output? (consider scanf)
Avoid vector, map. (use arrays/unordered_map)
What do your teammates think about your algorithm?

Memory limit exceeded:
```

What is the max amount of memory your algorithm should need?  
Are you clearing all data structures between test cases?

Data structures (2)

OrderStatisticTree.h782797, 16 lines

```
#include <bits/extc++.h>
using namespace __gnu_pbds;

template<class T>
using Tree = tree<T, null_type, less<T>, rb_tree_tag,
    tree_order_statistics_node_update>;

void example() {
    Tree<int> t, t2; t.insert(8);
    auto it = t.insert(10).first;
    assert(it == t.lower_bound(9));
    assert(t.order_of_key(10) == 1);
    assert(t.order_of_key(11) == 2);
    assert(*t.find_by_order(0) == 8);
    t.join(t2); // assuming T < T2 or T > T2, merge t2 into t
}
```

HashMap.hd77092, 7 lines

```
#include <bits/extc++.h>
// To use most bits rather than just the lowest ones:
struct chash { // large odd number for C
    const uint64_t C = 1l(4e18 * acos(0)) | 71;
    ll operator()(ll x) const { return __builtin_bswap64(x*C); }
};
__gnu_pbds::gp_hash_table<ll,int,chash> h({}, {}, {}, {}, {1<<16});
```

SegmentTree.h

```
Description: Zero-indexed max-tree. Bounds are inclusive to the left and exclusive to the right. Can be changed by modifying T, f and unit.
Time: O(log N)
Of4bdb, 19 lines

struct Tree {
    typedef int T;
    static constexpr T unit = INT_MIN;
    T f(T a, T b) { return max(a, b); } // (any associative fn)
    vector<T> s; int n;
    Tree(int n = 0, T def = unit) : s(2*n, def), n(n) {}
    void update(int pos, T val) {
        for (s[pos += n] = val; pos /= 2;)
            s[pos] = f(s[pos * 2], s[pos * 2 + 1]);
    }
    T query(int b, int e) { // query [b, e)
        T ra = unit, rb = unit;
        for (b += n, e += n; b < e; b /= 2, e /= 2) {
            if (b % 2) ra = f(ra, s[b++]);
            if (e % 2) rb = f(s[--e], rb);
        }
        return f(ra, rb);
    }
};
```

LazySegmentTree.h

**Description:** Segment tree with ability to add or set values of large intervals, and compute max of intervals. Can be changed to other things. Use with a bump allocator for better performance, and SmallPtr or implicit indices to save memory.

**Usage:** Node\* tr = new Node(v, 0, sz(v));

**Time:**  $\mathcal{O}(\log N)$ .

../various/BumpAllocator.h34ecf5, 50 lines

```
const int inf = 1e9;
struct Node {
    Node *l = 0, *r = 0;
    int lo, hi, mset = inf, madd = 0, val = -inf;
    Node(int lo,int hi):lo(lo),hi(hi){} // Large interval of -inf
    Node(vi& v, int lo, int hi) : lo(lo), hi(hi) {
        if (lo + 1 < hi) {
            int mid = lo + (hi - lo)/2;
            l = new Node(v, lo, mid); r = new Node(v, mid, hi);
            val = max(l->val, r->val);
        }
        else val = v[lo];
    }
    int query(int L, int R) {
        if (R <= lo || hi <= L) return -inf;
        if (L <= lo && hi <= R) return val;
        push();
        return max(l->query(L, R), r->query(L, R));
    }
    void set(int L, int R, int x) {
        if (R <= lo || hi <= L) return;
        if (L <= lo && hi <= R) mset = val = x, madd = 0;
        else {
            push(), l->set(L, R, x), r->set(L, R, x);
            val = max(l->val, r->val);
        }
    }
    void add(int L, int R, int x) {
        if (R <= lo || hi <= L) return;
        if (L <= lo && hi <= R) {
            if (mset != inf) mset += x;
            else madd += x;
            val += x;
        }
        else {
            push(), l->add(L, R, x), r->add(L, R, x);
            val = max(l->val, r->val);
        }
    }
    void push() {
        if (!l) {
            int mid = lo + (hi - lo)/2;
            l = new Node(lo, mid); r = new Node(mid, hi);
        }
        if (mset != inf)
            l->set(lo,hi,mset), r->set(lo,hi,mset), mset = inf;
        else if (madd)
            l->add(lo,hi,madd), r->add(lo,hi,madd), madd = 0;
    }
};
```

LiChaoTree.h

**Description:** Line range insetion and point queries, range insertion use upd2 in double log, upd1 is global insertion in single log, Note min conv use chains (i+j,Ai+Bj) instead of lines

**Time:**  $\mathcal{O}(\log N)$

fc9af8, 59 lines

```
int Div(int a){ return (a/2) - ((a < 0) && (a & 1));}
const ll INF = LLONG_MAX;
struct line{
    ll m = 0,c = INF;
    line(){};
```

```
    line(ll _m,ll _c): c(_c),m(_m){}
    ll eval(ll x){return m*x + c;}
};
struct node{
    line L0;
    node *lf = NULL,*rg = NULL;
    node(){}
    //for max just flip all <,>
    void upd1(int l,int r,line L1){
        if(l == r){
            if(L1.eval(l) < L0.eval(l))L0 = L1;
            return;
        }
        int tm = Div(l+r);
        if(L0.m < L1.m)swap(L0,L1);
        if(L1.eval(tm) < L0.eval(tm)){
            if(!lf)lf = new node();
            lf->upd1(l,tm,L0);
            L0 = L1;
        }else {
            if(!rg)rg = new node();
            rg->upd1(tm+1,r,L1);
        }
    }
    void upd2(int l,int r,int tl,int tr,line L1){
        if(l>r)return;
        if(l == tl && r == tr){
            upd1(l,r,L1);
            return;
        }
        if(!lf)lf = new node();
        if(!rg)rg = new node();
        int tm = Div(tl+tr);
        if(L0.m || L0.c != INF){
            lf->upd1(tl,tm,L0);
            rg->upd1(tm+1,tr,L0);
            L0 = line();
        }
        lf->upd2(l,min(r,tm),tl,tm,L1);
        rg->upd2(max(l,tm+1),r,tm+1,tr,L1);
    }
    ll query(int l,int r,int x,ll mn = INF){
        mn = min(mn,L0.eval(x));
        if(l == r){return mn;}
        int tm = Div(l+r);
        if(x <= tm){
            if(!lf)return mn;
            return lf->query(l,tm,x,mn);
        }else{
            if(!rg)return mn;
            return rg->query(tm+1,r,x,mn);
        }
    }
};
```

UnionFindRollback.h

**Description:** Disjoint-set data structure with undo. If undo is not needed, skip st, time() and rollback().

**Usage:** int t = uf.time(); ...; uf.rollback(t);

**Time:**  $\mathcal{O}(\log(N))$

de4ad0, 21 lines

```
struct RollbackUF {
    vi e; vector<pii> st;
    RollbackUF(int n) : e(n, -1) {}
    int size(int x) { return -e[find(x)]; }
    int find(int x) { return e[x] < 0 ? x : find(e[x]); }
    int time() { return sz(st); }
    void rollback(int t) {
        for (int i = time(); i --> t;)
```

```
        e[st[i].first] = st[i].second;
        st.resize(t);
    }
    bool join(int a, int b) {
        a = find(a), b = find(b);
        if (a == b) return false;
        if (e[a] > e[b]) swap(a, b);
        st.push_back({a, e[a]});
        st.push_back({b, e[b]});
        e[a] += e[b]; e[b] = a;
        return true;
    }
};
```

Matrix.h

**Description:** Basic operations on square matrices.

**Usage:** Matrix<int, 3> A;

A.d = {{{{1,2,3}}, {{4,5,6}}, {{7,8,9}}}};

array<int, 3> vec = {1,2,3};

vec = (A^N) \* vec;

6ab5db, 26 lines

```
template<class T, int N> struct Matrix {
    typedef Matrix M;
    array<array<T, N>, N> d{};
    M operator*(const M& m) const {
        M a;
        rep(i,0,N) rep(j,0,N)
            rep(k,0,N) a.d[i][j] += d[i][k]*m.d[k][j];
        return a;
    }
    array<T, N> operator*(const array<T, N>& vec) const {
        array<T, N> ret{};
        rep(i,0,N) rep(j,0,N) ret[i] += d[i][j] * vec[j];
        return ret;
    }
    M operator^(ll p) const {
        assert(p >= 0);
        M a, b(*this);
        rep(i,0,N) a.d[i][i] = 1;
        while (p) {
            if (p&1) a = a*b;
            b = b*b;
            p >>= 1;
        }
        return a;
    }
};
```

LineContainer.h

**Description:** Container where you can add lines of the form  $kx+m$ , and query maximum values at points  $x$ . Useful for dynamic programming (“convex hull trick”).

**Time:**  $\mathcal{O}(\log N)$

Sec1c7, 30 lines

```
struct Line {
    mutable ll k, m, p;
    bool operator<(const Line& o) const { return k < o.k; }
    bool operator<(ll x) const { return p < x; }
};

struct LineContainer : multiset<Line, less<>> {
    // (for doubles, use inf = 1/.0, div(a,b) = a/b)
    static const ll inf = LLONG_MAX;
    ll div(ll a, ll b) { // floored division
        return a / b - ((a ^ b) < 0 && a % b); }
    bool isect(iterator x, iterator y) {
        if (y == end()) return x->p = inf, 0;
        if (x->k == y->k) x->p = x->m > y->m ? inf : -inf;
        else x->p = div(y->m - x->m, x->k - y->k);
```

```
        return x->p >= y->p;
    }
    void add(ll k, ll m) {
        auto z = insert({k, m, 0}), y = z++, x = y;
        while (isect(y, z)) z = erase(z);
        if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
        while ((y = x) != begin() && (--x)->p >= y->p)
            isect(x, erase(y));
    }
    ll query(ll x) {
        assert(!empty());
        auto l = *lower_bound(x);
        return l.k * x + l.m;
    }
};
```

Treap.h  
**Description:** Segment tree but you can split and join segments If it is not used as implicit treap, you should compare val during split and merge  
**Time:**  $\mathcal{O}(\log N)$

```
struct Node {
    Node *l = 0, *r = 0;
    int val, y, c = 1;
    Node(int val) : val(val), y(rand()) {}

    //auxiliary info
    //int flip = 0, mn = 1e9, mmid = 0;
    void pull();
    void push();
    void apply();
};

int cnt(Node* n) { return n ? n->c : 0; }
void Node::apply() {}
void Node::push() {}
void Node::pull() {
    push();
    c = cnt(l) + cnt(r) + 1;
    //merge nodes here
}

pair<Node*, Node*> split(Node* n, int k) {
    if (!n) return {};
    n->push();
    if (cnt(n->l) >= k) { // "n->val >= k" for lower_bound(k)
        auto [L,R] = split(n->l, k);
        n->l = R;
        n->pull();
        return {L, n};
    } else {
        auto [L,R] = split(n->r, k - cnt(n->l) - 1); // and just "k"
        n->r = L;
        n->pull();
        return {n, R};
    }
}

Node* merge(Node* l, Node* r) {
    if (!l) return r;
    if (!r) return l;
    l->push();
    r->push();
    if (l->y > r->y) {
        l->r = merge(l->r, r);
        return l->pull(), l;
    } else {
        r->l = merge(l, r->l);
        return r->pull(), r;
    }
}
```

```
    }
}

Node* ins(Node* t, Node* n, int pos) {
    auto [l,r] = split(t, pos);
    return merge(merge(l, n), r);
}

// Example application: move the range [l, r) to index k
void move(Node*& t, int l, int r, int k) {
    Node *a, *b, *c;
    tie(a,b) = split(t, l); tie(b,c) = split(b, r - l);
    if (k <= l) t = merge(ins(a, b, k), c);
    else t = merge(a, ins(c, b, k - r));
}

}
```

FenwickTree.h  
**Description:** One-indexed fenwick tree  
**Time:** Both operations are  $\mathcal{O}(\log N)$ .

```
struct fenwick{
    int n;
    vector<ll>bit;
    void upd(int pos,ll val){
        for(int i=pos;i<=n;i+=i&(-i))bit[i]+=val;
    }
    ll query(int r){
        ll res = 0;
        for(int i=r;i>0;i-=i&(-i))res+=bit[i];
        return res;
    }
    int search(ll val){
        ll sum = 0,pos = 0;
        for(int i=19;i>0;i--){
            if(pos+(1<<i) < n && sum+bit[pos+(1<<i)]<val){
                pos+=(1<<i);
                sum+=bit[pos];
            }
        }
        return pos+1;
    }
    fenwick(int _n):n(_n),bit(vector<ll>(n+1,0)){}
};
```

FenwickTree2d.h  
**Description:** Computes sum of a[l..i][l..r], All points must be known beforehand, call add on all then init()

```
"FenwickTree.h"
struct fenwick2{
    int n;
    vector<fenwick>f;
    vv<int>pts;
    fenwick2(int _n):n(_n),f(n+1),pts(n+1){}
    void add(int x,int y){
        for(;x<=n;x += x&(-x))pts[x].push_back(y);
    }
    void init(){
        for(int i=1;i<=n;i++){
            sort(all(pts[i]));
            pts[i].resize(unique(all(pts[i])) - pts[i].begin());
            ;
            f[i] = fenwick(sz(pts[i]));
        }
    }
    void upd(int x,int y,int w){ //(x,y) must be added
        for(;x<=n;x += x&(-x))
            f[x].upd(lower_bound(all(pts[x]),y) - pts[x].begin()
                )+1,w);
    }
}
```

```
ll query(int x,int l,int r){
    ll ans = 0;
    for(;x>0;x -= x&(-x)){
        ans += f[x].query(lower_bound(all(pts[x]),l) - pts[x].begin());
        ans -= f[x].query(upper_bound(all(pts[x]),r) - pts[x].begin());
    }
    return ans;
}
};
```

RMQ.h  
**Description:** Range Minimum Queries on an array. Returns min(V[a], V[a + 1], ... V[b - 1]) in constant time.  
**Usage:** RMQ rmq(values);  
rmq.query(inclusive, exclusive);  
**Time:**  $\mathcal{O}(|V| \log |V| + Q)$

```
template<class T>
struct RMQ {
    vector<vector<T>> jmp;
    RMQ(const vector<T>& V) : jmp(1, V) {
        for (int pw = 1, k = 1; pw * 2 <= sz(V); pw *= 2, ++k) {
            jmp.emplace_back(sz(V) - pw * 2 + 1);
            rep(j,0,sz(jmp[k]))
                jmp[k][j] = min(jmp[k - 1][j], jmp[k - 1][j + pw]);
        }
    }
    T query(int a, int b) {
        assert(a < b); // or return inf if a == b
        int dep = 31 - __builtin_clz(b - a);
        return min(jmp[dep][a], jmp[dep][b - (1 << dep)]);
    }
};
```

MoQueries.h  
**Description:** Answer interval or tree path queries by finding an approximate TSP through the queries, and moving from one query to the next by adding/removing points at the ends. If values are on tree edges, change step to add/remove the edge (a,c) and remove the initial add call (but keep in).  
**Time:**  $\mathcal{O}(N\sqrt{Q})$

```
void add(int ind, int end) { ... } // add a[ind] (end = 0 or 1)
void del(int ind, int end) { ... } // remove a[ind]
int calc() { ... } // compute current answer

vi mo(vector<pii> Q) {
    int L = 0, R = 0, blk = 350; // ~N/sqrt(Q)
    vi s(sz(Q)), res = s;
    #define K(x) pii(x.first/blk, x.second ^ -(x.first/blk & 1))
    iota(all(s), 0);
    sort(all(s), [&](int s, int t){ return K(Q[s]) < K(Q[t]); });
    for (int qi : s) {
        pii q = Q[qi];
        while (L > q.first) add(--L, 0);
        while (R < q.second) add(R++, 1);
        while (L < q.first) del(L++, 0);
        while (R > q.second) del(--R, 1);
        res[qi] = calc();
    }
    return res;
}
```

```
vi moTree(vector<array<int, 2>> Q, vector<vi>& ed, int root=0){
    int N = sz(ed), pos[2] = {}, blk = 350; // ~N/sqrt(Q)
    vi s(sz(Q)), res = s, I(N), L(N), R(N), in(N), par(N);
    add(0, 0), in[0] = 1;
    auto dfs = [&](int x, int p, int dep, auto& f) -> void {
```

```
par[x] = p;
L[x] = N;
if (dep) I[x] = N++;
for (int y : ed[x]) if (y != p) f(y, x, !dep, f);
if (!dep) I[x] = N++;
R[x] = N;
};
dfs(root, -1, 0, dfs);
#define K(x) pii(I[x[0]] / blk, I[x[1]] ^ -(I[x[0]] / blk & 1))
iota(all(s), 0);
sort(all(s), [&](int s, int t){ return K(Q[s]) < K(Q[t]); });
for (int qi : s) rep(end,0,2) {
    int &a = pos[end], b = Q[qi][end], i = 0;
#define step(c) { if (in[c]) { del(a, end); in[a] = 0; } \
    else { add(c, end); in[c] = 1; } a = c; }
    while (!(L[b] <= L[a] && R[a] <= R[b]))
        I[i++] = b, b = par[b];
    while (a != b) step(par[a]);
    while (i--) step(I[i]);
    if (end) res[qi] = calc();
}
return res;
}
```

Linear Algebra (3)

3.1 Matrices

Determinant.h

**Description:** Calculates determinant of a matrix. Destroys the matrix.  
**Time:**  $O(N^3)$

---

bd5cec, 15 lines

```
double det(vector<vector<double>>& a) {
    int n = sz(a); double res = 1;
    rep(i,0,n) {
        int b = i;
        rep(j,i+1,n) if (fabs(a[j][i]) > fabs(a[b][i])) b = j;
        if (i != b) swap(a[i], a[b]), res *= -1;
        res *= a[i][i];
        if (res == 0) return 0;
        rep(j,i+1,n) {
            double v = a[j][i] / a[i][i];
            if (v != 0) rep(k,i+1,n) a[j][k] -= v * a[i][k];
        }
    }
    return res;
}
```

IntDeterminant.h

**Description:** Calculates determinant using modular arithmetics. Modulos can also be removed to get a pure-integer version.  
**Time:**  $O(N^3)$

---

3313dc, 18 lines

```
const ll mod = 12345;
ll det(vector<vector<ll>>& a) {
    int n = sz(a); ll ans = 1;
    rep(i,0,n) {
        rep(j,i+1,n) {
            while (a[j][i] != 0) { // gcd step
                ll t = a[i][i] / a[j][i];
                if (t) rep(k,i,n)
                    a[i][k] = (a[i][k] - a[j][k] * t) % mod;
                swap(a[i], a[j]);
                ans *= -1;
            }
        }
    }
    ans = ans * a[i][i] % mod;
    if (!ans) return 0;
}
```

```
}
return (ans + mod) % mod;
}

SolveLinear.h
Description: Solves  $A * x = b$ . If there are multiple solutions, an arbitrary one is returned. Returns rank, or -1 if no solutions. Data in  $A$  and  $b$  is lost.
Time:  $O(n^2m)$ 
```

---

44c9ab, 38 lines

```
typedef vector<double> vd;
const double eps = 1e-12;

int solveLinear(vector<vd>& A, vd& b, vd& x) {
    int n = sz(A), m = sz(x), rank = 0, br, bc;
    if (n) assert(sz(A[0]) == m);
    vi col(m); iota(all(col), 0);

    rep(i,0,n) {
        double v, bv = 0;
        rep(r,i,n) rep(c,i,m)
            if ((v = fabs(A[r][c])) > bv)
                br = r, bc = c, bv = v;
        if (bv <= eps) {
            rep(j,i,n) if (fabs(b[j]) > eps) return -1;
            break;
        }
        swap(A[i], A[br]);
        swap(b[i], b[br]);
        swap(col[i], col[bc]);
        rep(j,0,n) swap(A[j][i], A[j][bc]);
        bv = 1/A[i][i];
        rep(j,i+1,n) {
            double fac = A[j][i] * bv;
            b[j] -= fac * b[i];
            rep(k,i+1,m) A[j][k] -= fac*A[i][k];
        }
        rank++;
    }

    x.assign(m, 0);
    for (int i = rank; i--;) {
        b[i] /= A[i][i];
        x[col[i]] = b[i];
        rep(j,0,i) b[j] -= A[j][i] * b[i];
    }
    return rank; // (multiple solutions if rank < m)
}
```

SolveLinear2.h

**Description:** To get all uniquely determined values of  $x$  back from SolveLinear, make the following changes:

---

08e495, 7 lines

```
"SolveLinear.h"
rep(j,0,n) if (j != i) // instead of rep(j,i+1,n)
// ... then at the end:
x.assign(m, undefined);
rep(i,0,rank) {
    rep(j,rank,m) if (fabs(A[i][j]) > eps) goto fail;
    x[col[i]] = b[i] / A[i][i];
    fail:; }

SolveLinearBinary.h
Description: Solves  $Ax = b$  over  $\mathbb{F}_2$ . If there are multiple solutions, one is returned arbitrarily. Returns rank, or -1 if no solutions. Destroys  $A$  and  $b$ .
Time:  $O(n^2m)$ 
```

---

fa2d7a, 34 lines

```
typedef bitset<1000> bs;

int solveLinear(vector<bs>& A, vi& b, bs& x, int m) {
    int n = sz(A), rank = 0, br;
```

```
assert(m <= sz(x));
vi col(m); iota(all(col), 0);
rep(i,0,n) {
    for (br=i; br<n; ++br) if (A[br].any()) break;
    if (br == n) {
        rep(j,i,n) if(b[j]) return -1;
        break;
    }
    int bc = (int)A[br]._Find_next(i-1);
    swap(A[i], A[br]);
    swap(b[i], b[br]);
    swap(col[i], col[bc]);
    rep(j,0,n) if (A[j][i] != A[j][bc]) {
        A[j].flip(i); A[j].flip(bc);
    }
    rep(j,i+1,n) if (A[j][i]) {
        b[j] ^= b[i];
        A[j] ^= A[i];
    }
    rank++;
}

x = bs();
for (int i = rank; i--;) {
    if (!b[i]) continue;
    x[col[i]] = 1;
    rep(j,0,i) b[j] ^= A[j][i];
}
return rank; // (multiple solutions if rank < m)
}
```

MatrixInverse.h

**Description:** Invert matrix  $A$ . Returns rank; result is stored in  $A$  unless singular (rank < n). Can easily be extended to prime moduli; for prime powers, repeatedly set  $A^{-1} = A^{-1}(2I - AA^{-1}) \pmod{p^k}$  where  $A^{-1}$  starts as the inverse of  $A \pmod p$ , and  $k$  is doubled in each step.  
**Time:**  $O(n^3)$

---

ebfff6, 35 lines

```
int matInv(vector<vector<double>>& A) {
    int n = sz(A); vi col(n);
    vector<vector<double>> tmp(n, vector<double>(n));
    rep(i,0,n) tmp[i][i] = 1, col[i] = i;

    rep(i,0,n) {
        int r = i, c = i;
        rep(j,i,n) rep(k,i,n)
            if (fabs(A[j][k]) > fabs(A[r][c]))
                r = j, c = k;
        if (fabs(A[r][c]) < 1e-12) return i;
        A[i].swap(A[r]); tmp[i].swap(tmp[r]);
        rep(j,0,n)
            swap(A[j][i], A[j][c]), swap(tmp[j][i], tmp[j][c]);
        swap(col[i], col[c]);
        double v = A[i][i];
        rep(j,i+1,n) {
            double f = A[j][i] / v;
            A[j][i] = 0;
            rep(k,i+1,n) A[j][k] -= f*A[i][k];
            rep(k,0,n) tmp[j][k] -= f*tmp[i][k];
        }
        rep(j,i+1,n) A[i][j] /= v;
        rep(j,0,n) tmp[i][j] /= v;
        A[i][i] = 1;
    }

    for (int i = n-1; i > 0; --i) rep(j,0,i) {
        double v = A[j][i];
        rep(k,0,n) tmp[j][k] -= v*tmp[i][k];
    }
}
```

```
rep(i,0,n) rep(j,0,n) A[col[i]][col[j]] = tmp[i][j];
return n;
}
```

Polynomial (4)

4.1 Poly manipulation

**Poly.h**  
**Description:** supports inv(P),ln(P),exp(P) all in nlogn. Not optimized, might have TLE issues  
**Time:**  $\mathcal{O}(N \log N)$ , where  $N = deg(P)$

```
"NumberTheoreticTransform.h" a9c994, 62 lines
ll inv(ll a) {
    return a <= 1 ? a : mod - (mod/a) * inv(mod % a) % mod;
}
using poly = vl;
poly rsz(const poly& p, int x) {
    if (x <= sz(p)) return poly(begin(p),begin(p)+x);
    poly q = p; q.resize(x); return q;
}
poly& operator+=(poly& p, const poly& q) {
    p = rsz(p,max(sz(p),sz(q)));
    rep(i, 0,sz(p)) {p[i] += q[i]; if(p[i] >= mod)p[i]-=mod;}
    return p;
}
poly& operator-=(poly& p, const poly& q) {
    p = rsz(p,max(sz(p),sz(q)));
    rep(i, 0,sz(p)) {p[i] += mod-q[i];if(p[i] >= mod)p[i]-=mod;
    }
    return p;
}
poly& operator*=(poly& p, const int &k) {
    for(auto &x:p)x = (ll)x*k%mod;
    return p;
}

poly operator+(poly p, const poly& q) { return p += q; }
poly operator-(poly p, const poly& q) { return p -= q; }
poly operator-(poly p) { for(auto &x:p)x = x*(mod-1)%mod;
    return p; }
poly operator*(poly p, const int& k) { return p *= k; }
poly operator*(const int& k, const poly& p) { return p*k; }
poly operator*(poly p, const poly& q){return conv(p,q);}

poly dif(const poly& p) { // differentiate
    poly res;
    rep(i, 1, sz(p)) res.pb((ll)i*p[i]%mod);
    return res;
}
poly integ(poly p){
    static poly invs{0,1};
    poly res(sz(p)+1);
    rep(i, sz(invs), sz(p)+1) invs.pb(mod - (ll)mod/i * invs[
        mod%i] % mod);
    rep(i, 0, sz(p)) res[i+1] = (ll)p[i]*invs[i+1] % mod;
    return res;
}
poly inv(poly A, int n) { //Q_{-{k+1}} = Q_k(2-P*Q_k)
    poly B{inv(A[0])};
    for (int x = 2; x/2 < n; x<=1) B = 2*B-rsz(conv(rsz(A,x),
        conv(B,B)),x);
    return rsz(B,n);
}

poly ln(poly p, int n){
    poly q = rsz(dif(p) * inv(p,n-1),n-1);
```

```
return rsz(integ(q),n);
}

poly exp(poly A, int n) { assert(A[0] == 0);
    poly B{1}, IB{1};
    for (int x = 1; x < n; x<=1) {
        IB = 2*IB-rsz(conv(B,conv(IB,IB)),x);
        poly Q = dif(rsz(A,x)); Q += rsz(conv(IB,dif(B)-conv(B,Q)),
            2*x-1);
        B = B+rsz(conv(B,rsz(A,2*x)-integ(Q)),2*x);
    }
    return rsz(B,n);
}
```

Polynomial.h

```
struct Poly {
    vector<double> a;
    double operator() (double x) const {
        double val = 0;
        for (int i = sz(a); i--;) (val += x) += a[i];
        return val;
    }
    void diff() {
        rep(i,1,sz(a)) a[i-1] = i*a[i];
        a.pop_back();
    }
    void divroot(double x0) {
        double b = a.back(), c; a.back() = 0;
        for(int i=sz(a)-1; i--;) c = a[i], a[i] = a[i+1]*x0+b, b=c;
        a.pop_back();
    }
};
```

PolyRoots.h

**Description:** Finds the real roots to a polynomial.  
**Usage:** polyRoots({{2,-3,1}},-1e9,1e9) // solve x^2-3x+2 = 0  
**Time:**  $\mathcal{O}(n^2 \log(1/\epsilon))$

```
"Polynomial.h" b00bfe, 23 lines
vector<double> polyRoots(Poly p, double xmin, double xmax) {
    if (sz(p.a) == 2) { return {-p.a[0]/p.a[1]}; }
    vector<double> ret;
    Poly der = p;
    der.diff();
    auto dr = polyRoots(der, xmin, xmax);
    dr.push_back(xmin-1);
    dr.push_back(xmax+1);
    sort(all(dr));
    rep(i,0,sz(dr)-1) {
        double l = dr[i], h = dr[i+1];
        bool sign = p(l) > 0;
        if (sign ^ (p(h) > 0)) {
            rep(it,0,60) { // while (h - l > 1e-8)
                double m = (l + h) / 2, f = p(m);
                if ((f <= 0) ^ sign) l = m;
                else h = m;
            }
            ret.push_back((l + h) / 2);
        }
    }
    return ret;
}
```

PolyInterpolate.h

**Description:** Given  $n$  points  $(x[i], y[i])$ , computes an  $n-1$ -degree polynomial  $p$  that passes through them:  $p(x) = a[0] * x^0 + ... + a[n-1] * x^{n-1}$ . For numerical precision, pick  $x[k] = c * \cos(k/(n-1) * \pi), k = 0...n-1$ .  
**Time:**  $\mathcal{O}(n^2)$

```
typedef vector<double> vd;
vd interpolate(vd x, vd y, int n) {
    vd res(n), temp(n);
    rep(k,0,n-1) rep(i,k+1,n)
        y[i] = (y[i] - y[k]) / (x[i] - x[k]);
    double last = 0; temp[0] = 1;
    rep(k,0,n) rep(i,0,n) {
        res[i] += y[k] * temp[i];
        swap(last, temp[i]);
        temp[i] -= last * x[k];
    }
    return res;
}
```

BerlekampMassey.h

**Description:** Recovers any  $n$ -order linear recurrence relation from the first  $2n$  terms of the recurrence. Useful for guessing linear recurrences after brute-forcing the first terms. Should work on any field, but numerical stability for floats is not guaranteed. Output will have size  $\leq n$ .  
**Usage:** berlekampMassey({0, 1, 1, 3, 5, 11}) // {1, 2}  
**Time:**  $\mathcal{O}(N^2)$

```
"../number-theory/ModPow.h" 96548b, 20 lines
vector<ll> berlekampMassey(vector<ll> s) {
    int n = sz(s), L = 0, m = 0;
    vector<ll> C(n), B(n), T;
    C[0] = B[0] = 1;

    ll b = 1;
    rep(i,0,n) { ++m;
        ll d = s[i] % mod;
        rep(j,1,L+1) d = (d + C[j] * s[i - j]) % mod;
        if (!d) continue;
        T = C; ll coef = d * modpow(b, mod-2) % mod;
        rep(j,m,n) C[j] = (C[j] - coef * B[j - m]) % mod;
        if (2 * L > i) continue;
        L = i + 1 - L; B = T; b = d; m = 0;
    }

    C.resize(L + 1); C.erase(C.begin());
    for (ll& x : C) x = (mod - x) % mod;
    return C;
}
```

LinearRecurrence.h

**Description:** Generates the  $k$ 'th term of an  $n$ -order linear recurrence  $S[i] = \sum_j S[i-j-1]tr[j]$ , given  $S[0... \geq n-1]$  and  $tr[0...n-1]$ . Faster than matrix multiplication. Useful together with Berlekamp-Massey.  
**Usage:** linearRec({0, 1}, {1, 1}, k) //  $k$ 'th Fibonacci number  
**Time:**  $\mathcal{O}(n^2 \log k)$

```
typedef vector<ll> Poly;
ll linearRec(Poly S, Poly tr, ll k) {
    int n = sz(tr);

    auto combine = [&](Poly a, Poly b) {
        Poly res(n * 2 + 1);
        rep(i,0,n+1) rep(j,0,n+1)
            res[i + j] = (res[i + j] + a[i] * b[j]) % mod;
        for (int i = 2 * n; i > n; --i) rep(j,0,n)
            res[i - 1 - j] = (res[i - 1 - j] + res[i] * tr[j]) % mod;
        res.resize(n + 1);
        return res;
    };

    Poly pol(n + 1), e(pol);
    pol[0] = e[1] = 1;

    for (++k; k; k /= 2) {
```

```
    if (k % 2) pol = combine(pol, e);
    e = combine(e, e);
}

ll res = 0;
rep(i,0,n) res = (res + pol[i + 1] * S[i]) % mod;
return res;
}
```

## 4.2 Optimization

### GoldenSectionSearch.h

**Description:** Finds the argument minimizing the function  $f$  in the interval  $[a, b]$  assuming  $f$  is unimodal on the interval, i.e. has only one local minimum and no local maximum. The maximum error in the result is  $\epsilon$ *ps*. Works equally well for maximization with a small change in the code. See Ternary-Search.h in the Various chapter for a discrete version.

**Usage:** double func(double x) { return 4+x+.3\*x\*x; }  
double xmin = gss(-1000,1000,func);  
**Time:**  $\mathcal{O}(\log((b-a)/\epsilon))$

31d45b, 14 lines

```
double gss(double a, double b, double (*f)(double)) {
    double r = (sqrt(5)-1)/2, eps = 1e-7;
    double x1 = b - r*(b-a), x2 = a + r*(b-a);
    double f1 = f(x1), f2 = f(x2);
    while (b-a > eps)
        if (f1 < f2) { //change to > to find maximum
            b = x2; x2 = x1; f2 = f1;
            x1 = b - r*(b-a); f1 = f(x1);
        } else {
            a = x1; x1 = x2; f1 = f2;
            x2 = a + r*(b-a); f2 = f(x2);
        }
    return a;
}
```

### Simplex.h

**Description:** Solves a general linear maximization problem: maximize  $c^T x$  subject to  $Ax \leq b, x \geq 0$ . Returns -inf if there is no solution, inf if there are arbitrarily good solutions, or the maximum value of  $c^T x$  otherwise. The input vector is set to an optimal  $x$  (or in the unbounded case, an arbitrary solution fulfilling the constraints). Numerical stability is not guaranteed. For better performance, define variables such that  $x = 0$  is viable.

**Usage:** vvd A = {{1,-1}, {-1,1}, {-1,-2}};  
vd b = {1,1,-4}, c = {-1,-1}, x;  
T val = LPSolver(A, b, c).solve(x);  
**Time:**  $\mathcal{O}(NM * \#pivots)$ , where a pivot may be e.g. an edge relaxation.  $\mathcal{O}(2^n)$  in the general case.

aa8530, 68 lines

```
typedef double T; // long double, Rational, double + mod<P>...
typedef vector<T> vd;
typedef vector<vd> vvd;
```

```
const T eps = 1e-8, inf = 1/.0;
#define MP make_pair
#define ltj(X) if(s == -1 || MP(X[j],N[j]) < MP(X[s],N[s])) s=j
```

```
struct LPSolver {
    int m, n;
    vi N, B;
    vvd D;

    LPSolver(const vvd& A, const vd& b, const vd& c) :
        m(sz(b)), n(sz(c)), N(n+1), B(m), D(m+2, vd(n+2)) {
        rep(i,0,m) rep(j,0,n) D[i][j] = A[i][j];
        rep(i,0,m) { B[i] = n+i; D[i][n] = -1; D[i][n+1] = b[i]; }
        rep(j,0,n) { N[j] = j; D[m][j] = -c[j]; }
        N[n] = -1; D[m+1][n] = 1;
    }
}
```

```
void pivot(int r, int s) {
    T *a = D[r].data(), inv = 1 / a[s];
    rep(i,0,m+2) if (i != r && abs(D[i][s]) > eps) {
        T *b = D[i].data(), inv2 = b[s] * inv;
        rep(j,0,n+2) b[j] -= a[j] * inv2;
        b[s] = a[s] * inv2;
    }
    rep(j,0,n+2) if (j != s) D[r][j] *= inv;
    rep(i,0,m+2) if (i != r) D[i][s] *= -inv;
    D[r][s] = inv;
    swap(B[r], N[s]);
}
```

```
bool simplex(int phase) {
    int x = m + phase - 1;
    for (;;) {
        int s = -1;
        rep(j,0,n+1) if (N[j] != -phase) ltj(D[x]);
        if (D[x][s] >= -eps) return true;
        int r = -1;
        rep(i,0,m) {
            if (D[i][s] <= eps) continue;
            if (r == -1 || MP(D[i][n+1] / D[i][s], B[i])
                        < MP(D[r][n+1] / D[r][s], B[r])) r = i;
        }
        if (r == -1) return false;
        pivot(r, s);
    }
}
```

```
T solve(vd &x) {
    int r = 0;
    rep(i,1,m) if (D[i][n+1] < D[r][n+1]) r = i;
    if (D[r][n+1] < -eps) {
        pivot(r, n);
        if (!simplex(2) || D[m+1][n+1] < -eps) return -inf;
        rep(i,0,m) if (B[i] == -1) {
            int s = 0;
            rep(j,1,n+1) ltj(D[i]);
            pivot(i, s);
        }
    }
    bool ok = simplex(1); x = vd(n);
    rep(i,0,m) if (B[i] < n) x[B[i]] = D[i][n+1];
    return ok ? D[m][n+1] : inf;
}
```

## 4.3 Fourier transforms

### FastFourierTransform.h

**Description:** fft(a) computes  $\hat{f}(k) = \sum_x a[x] \exp(2\pi i \cdot kx/N)$  for all  $k$ .  $N$  must be a power of 2. Useful for convolution: conv(a, b) = c, where  $c[x] = \sum a[i]b[x-i]$ . For convolution of complex numbers or more than two vectors: FFT, multiply pointwise, divide by n, reverse(start+1, end), FFT back. Rounding is safe if  $(\sum a_i^2 + \sum b_i^2) \log_2 N < 9 \cdot 10^{14}$  (in practice  $10^{16}$ ; higher for random inputs). Otherwise, use NTT/FFTMod.  
**Time:**  $\mathcal{O}(N \log N)$  with  $N = |A| + |B|$  ( $\sim 1$ s for  $N = 2^{22}$ )

00ced6, 35 lines

```
typedef complex<double> C;
typedef vector<double> vd;
void fft(vector<C>& a) {
    int n = sz(a), L = 31 - __builtin_clz(n);
    static vector<complex<long double>> R(2, 1);
    static vector<C> rt(2, 1); // (^ 10% faster if double)
    for (static int k = 2; k < n; k *= 2) {
        R.resize(n); rt.resize(n);
        auto x = polar(1.0L, acos(-1.0L) / k);
        rep(i,k,2*k) rt[i] = R[i] = i&1 ? R[i/2] * x : R[i/2];
    }
```

```
vi rev(n);
rep(i,0,n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
rep(i,0,n) if (i < rev[i]) swap(a[i], a[rev[i]]);
for (int k = 1; k < n; k *= 2)
    for (int i = 0; i < n; i += 2 * k) rep(j,0,k) {
        C z = rt[j+k] * a[i+j+k]; // (25% faster if hand-rolled)
        a[i + j + k] = a[i + j] - z;
        a[i + j] += z;
    }
}

vd conv(const vd& a, const vd& b) {
    if (a.empty() || b.empty()) return {};
    vd res(sz(a) + sz(b) - 1);
    int L = 32 - __builtin_clz(sz(res)), n = 1 << L;
    vector<C> in(n), out(n);
    copy(all(a), begin(in));
    rep(i,0,sz(b)) in[i].imag(b[i]);
    fft(in);
    for (C& x : in) x *= x;
    rep(i,0,n) out[i] = in[-i & (n - 1)] - conj(in[i]);
    fft(out);
    rep(i,0,sz(res)) res[i] = imag(out[i]) / (4 * n);
    return res;
}
```

### FastFourierTransformMod.h

**Description:** Higher precision FFT, can be used for convolutions modulo arbitrary integers as long as  $N \log_2 N \cdot \text{mod} < 8.6 \cdot 10^{14}$  (in practice  $10^{16}$  or higher). Inputs must be in  $[0, \text{mod})$ .

**Time:**  $\mathcal{O}(N \log N)$ , where  $N = |A| + |B|$  (twice as slow as NTT or FFT)

b82773, 22 lines

```
typedef vector<ll> vl;
template<int M> vl convMod(const vl &a, const vl &b) {
    if (a.empty() || b.empty()) return {};
    vl res(sz(a) + sz(b) - 1);
    int B=32-__builtin_clz(sz(res)), n=1<<B, cut=int(sqrt(M));
    vector<C> L(n), R(n), outs(n), outl(n);
    rep(i,0,sz(a)) L[i] = C((int)a[i] / cut, (int)a[i] % cut);
    rep(i,0,sz(b)) R[i] = C((int)b[i] / cut, (int)b[i] % cut);
    fft(L), fft(R);
    rep(i,0,n) {
        int j = -i & (n - 1);
        outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
        outs[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / 1i;
    }
    fft(outl), fft(outs);
    rep(i,0,sz(res)) {
        ll av = ll(real(outl[i])+.5), cv = ll(imag(outs[i])+.5);
        ll bv = ll(imag(outl[i])+.5) + ll(real(outs[i])+.5);
        res[i] = ((av % M * cut + bv) % M * cut + cv) % M;
    }
    return res;
}
```

### NumberTheoreticTransform.h

**Description:** ntt(a) computes  $\hat{f}(k) = \sum_x a[x]g^{xk}$  for all  $k$ , where  $g = \text{root}^{(\text{mod}-1)/N}$ .  $N$  must be a power of 2. Useful for convolution modulo specific nice primes of the form  $2^a b + 1$ , where the convolution result has size at most  $2^a$ . For arbitrary modulo, see FFTMod. conv(a, b) = c, where  $c[x] = \sum a[i]b[x-i]$ . For manual convolution: NTT the inputs, multiply pointwise, divide by n, reverse(start+1, end), NTT back. Inputs must be in  $[0, \text{mod})$ .

**Time:**  $\mathcal{O}(N \log N)$

ced03d, 35 lines

```
.../number-theory/ModPow.h"
const ll mod = (119 << 23) + 1, root = 62; // = 998244353
// For p < 2^30 there is also e.g. 5 << 25, 7 << 26, 479 << 21
// and 483 << 21 (same root). The last two are > 10^9.
typedef vector<ll> vl;
```

```
void ntt(vl &a) {
    int n = sz(a), L = 31 - __builtin_clz(n);
    static vl rt(2, 1);
    for (static int k = 2, s = 2; k < n; k *= 2, s++) {
        rt.resize(n);
        ll z[] = {1, modpow(root, mod >> s)};
        rep(i,k,2*k) rt[i] = rt[i / 2] * z[i & 1] % mod;
    }
    vi rev(n);
    rep(i,0,n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
    rep(i,0,n) if (i < rev[i]) swap(a[i], a[rev[i]]);
    for (int k = 1; k < n; k *= 2)
        for (int i = 0; i < n; i += 2 * k) rep(j,0,k) {
            ll z = rt[j + k] * a[i + j + k] % mod, &ai = a[i + j];
            a[i + j + k] = ai - z + (z > ai ? mod : 0);
            ai += (ai + z >= mod ? z - mod : z);
        }
}

vl conv(const vl &a, const vl &b) {
    if (a.empty() || b.empty()) return {};
    int s = sz(a) + sz(b) - 1, B = 32 - __builtin_clz(s),
        n = 1 << B;
    int inv = modpow(n, mod - 2);
    vl L(a), R(b), out(n);
    L.resize(n), R.resize(n);
    ntt(L), ntt(R);
    rep(i,0,n)
        out[-i & (n - 1)] = (ll)L[i] * R[i] % mod * inv % mod;
    ntt(out);
    return {out.begin(), out.begin() + s};
}
```

### FastSubsetTransform.h

**Description:** Transform to a basis with fast convolutions of the form  $c[z] = \sum_{z=x\oplus y} a[x] \cdot b[y]$ , where  $\oplus$  is one of AND, OR, XOR. The size of  $a$  must be a power of two.  
**Time:**  $\mathcal{O}(N \log N)$

```
void FST(vi& a, bool inv) {
    for (int n = sz(a), step = 1; step < n; step *= 2) {
        for (int i = 0; i < n; i += 2 * step) rep(j,i,step) {
            int &u = a[j], &v = a[j + step]; tie(u, v) =
                inv ? pii(v - u, u) : pii(v, u + v); // AND
            inv ? pii(v, u - v) : pii(u + v, u); // OR
            pii(u + v, u - v); // XOR
        }
    }
    if (inv) for (int& x : a) x /= sz(a); // XOR only
}

vi conv(vi a, vi b) {
    FST(a, 0); FST(b, 0);
    rep(i,0,sz(a)) a[i] *= b[i];
    FST(a, 1); return a;
}
```

### Zeta.h

**Description:** 1-index zeta transform for gcd convolution and lcm convolution. There should be another function(Mobius) that reverse the change after multiply every entry  
**Usage:** zeta(a), zeta(b);  
rep(i,0,n) c[i]=a[i]\*b[i];  
reverseZeta(c);

```
"../number-theory/Sieve.h"
7401ac, 7 lines

void zeta(vi& v) {
    int n = sz(v)-1;
    for (int p : ps) {
        for (int i = 1; i*p <= n; ++i) v[i * p] += v[i]; // divisor
            zeta, for lcm
    }
```

```
        for (int i = n/p; i; --i) v[i] += v[i * p]; // multiple
            zeta, for gcd
    }
}
```

## Number theory (5)

### 5.1 Modular arithmetic

#### ModularArithmetic.h

**Description:** Operators for modular arithmetic. You need to set mod to some number first and then you can use the structure.

```
"euclid.h"
35bfea, 18 lines

const ll mod = 17; // change to something else
struct Mod {
    ll x;
    Mod(ll xx) : x(xx) {}
    Mod operator+(Mod b) { return Mod((x + b.x) % mod); }
    Mod operator-(Mod b) { return Mod((x - b.x + mod) % mod); }
    Mod operator*(Mod b) { return Mod((x * b.x) % mod); }
    Mod operator/(Mod b) { return *this * invert(b); }
    Mod invert(Mod a) {
        ll x, y, g = euclid(a.x, mod, x, y);
        assert(g == 1); return Mod((x + mod) % mod);
    }
    Mod operator^(ll e) {
        if (!e) return Mod(1);
        Mod r = *this ^ (e / 2); r = r * r;
        return e&1 ? *this * r : r;
    }
};
```

#### ModInverse.h

**Description:** Pre-computation of modular inverses. Assumes  $\text{LIM} \leq \text{mod}$  and that mod is a prime.

```
const ll mod = 1000000007, LIM = 200000;
ll* inv = new ll[LIM] - 1; inv[1] = 1;
rep(i,2,LIM) inv[i] = mod - (mod / i) * inv[mod % i] % mod;
```

#### ModPow.h

```
const ll mod = 1000000007; // faster if const

ll modpow(ll b, ll e) {
    ll ans = 1;
    for (; e; b = b * b % mod, e /= 2)
        if (e & 1) ans = ans * b % mod;
    return ans;
}
```

#### ModLog.h

**Description:** Returns the smallest  $x > 0$  s.t.  $a^x = b \pmod m$ , or  $-1$  if no such  $x$  exists. modLog(a,1,m) can be used to calculate the order of  $a$ .  
**Time:**  $\mathcal{O}(\sqrt{m})$

```
ll modLog(ll a, ll b, ll m) {
    ll n = (ll) sqrt(m) + 1, e = 1, f = 1, j = 1;
    unordered_map<ll, ll> A;
    while (j <= n && (e = f = e * a % m) != b % m)
        A[e * b % m] = j++;
    if (e == b % m) return j;
    if (__gcd(m, e) == __gcd(m, b))
        rep(i,2,n+2) if (A.count(e = e * f % m))
            return n * i - A[e];
    return -1;
}
```

#### ModSum.h

**Description:** Sums of mod'ed arithmetic progressions.  
 $\text{modsum}(to, c, k, m) = \sum_{i=0}^{to-1} (ki + c) \% m$ . divsum is similar but for floored division.  
**Time:**  $\log(m)$ , with a large constant.

```
typedef unsigned long long ull;
ull sumsq(ull to) { return to / 2 * ((to-1) | 1); }
```

```
ull divsum(ull to, ull c, ull k, ull m) {
    ull res = k / m * sumsq(to) + c / m * to;
    k %= m; c %= m;
    if (!k) return res;
    ull to2 = (to * k + c) / m;
    return res + (to - 1) * to2 - divsum(to2, m-1 - c, m, k);
}
```

```
ll modsum(ull to, ll c, ll k, ll m) {
    c = ((c % m) + m) % m;
    k = ((k % m) + m) % m;
    return to * c + k * sumsq(to) - m * divsum(to, c, k, m);
}
```

#### ModMulLL.h

**Description:** Calculate  $a \cdot b \pmod c$  (or  $a^b \pmod c$ ) for  $0 \leq a, b \leq c \leq 7.2 \cdot 10^{18}$ .  
**Time:**  $\mathcal{O}(1)$  for modmul,  $\mathcal{O}(\log b)$  for modpow

```
typedef unsigned long long ull;
ull modmul(ull a, ull b, ull M) {
    ll ret = a * b - M * ull(1.L / M * a * b);
    return ret + M * (ret < 0) - M * (ret >= (ll)M);
}

ull modpow(ull b, ull e, ull mod) {
    ull ans = 1;
    for (; e; b = modmul(b, b, mod), e /= 2)
        if (e & 1) ans = modmul(ans, b, mod);
    return ans;
}
```

#### ModSqrt.h

**Description:** Tonelli-Shanks algorithm for modular square roots. Finds  $x$  s.t.  $x^2 = a \pmod p$  ( $-x$  gives the other solution).  
**Time:**  $\mathcal{O}(\log^2 p)$  worst case,  $\mathcal{O}(\log p)$  for most  $p$

```
"ModPow.h"
19a793, 24 lines

ll sqrt(ll a, ll p) {
    a %= p; if (a < 0) a += p;
    if (a == 0) return 0;
    assert(modpow(a, (p-1)/2, p) == 1); // else no solution
    if (p % 4 == 3) return modpow(a, (p+1)/4, p);
    // a^(n+3)/8 or 2^(n+3)/8 * 2^(n-1)/4 works if p % 8 == 5
    ll s = p - 1, n = 2;
    int r = 0, m;
    while (s % 2 == 0)
        ++r, s /= 2;
    while (modpow(n, (p - 1) / 2, p) != p - 1) ++n;
    ll x = modpow(a, (s + 1) / 2, p);
    ll b = modpow(a, s, p), g = modpow(n, s, p);
    for (; r = m) {
        ll t = b;
        for (m = 0; m < r && t != 1; ++m)
            t = t * t % p;
        if (m == 0) return x;
        ll gs = modpow(g, 1LL << (r - m - 1), p);
        g = gs * gs % p;
        x = x * gs % p;
        b = b * g % p;
    }
}
```

PrimitiveRoot.h

**Description:** modified rho if p is not prime.  
for number <= 1e12, it's simpler to sieve primes and factorize the number yourself.

**Time:** Unknown but good! need to factorize p-1  
"./contest/random.h", "ModMulLL.h";, "Factor.h", "MillerRabin.h" a9b2e3, 14 lines

```
ll primitive_root(ll p) {
    if (p == 2) return 1;
    ull rho = p-1;
    auto pf = factor(rho); // should be one of vi, v<ll>, v<ull
    >
    sort(all(pf)); pf.erase(unique(all(pf)), pf.end());
    while ( 1 ) {
        ll g = rnd(1,p);
        bool ok = 1;
        for (auto f : pf) {
            if (modpow(g, rho/f, p) == 1) { ok = 0; break; }
        }
        if (ok) return g;
    }
}
```

5.2 Primality

Sieve.h

**Description:** generating all primes smaller than LIM, minp[i] store minimum prime factor of i, could be use for factorize a number.

**Time:**  $\mathcal{O}(LIM)$  e76c38, 13 lines

```
const int LIM=1e6+5;
int minp[LIM];
void ps;
void sieve(){
    minp[1] = 1;
    rep(i, 2, LIM) {
        if ( !minp[i] ) ps.pb ( minp[i] = i );
        for ( int p: ps ) {
            if ( p > minp[i] || i*1LL*p >= LIM ) break;
            minp[i*p] = p;
        }
    }
}
```

MillerRabin.h

**Description:** Deterministic Miller-Rabin primality test. Guaranteed to work for numbers up to  $7 \cdot 10^{18}$ ; for larger numbers, use Python and extend A randomly.

**Time:** 7 times the complexity of  $a^b \bmod c$ .  
"ModMulLL.h" 60dcd1, 12 lines

```
bool isPrime(ull n) {
    if (n < 2 || n % 6 % 4 != 1) return (n | 1) == 3;
    ull A[] = {2, 325, 9375, 28178, 450775, 9780504, 1795265022},
    s = __builtin_ctzll(n-1), d = n >> s;
    for (ull a : A) { // ^ count trailing zeroes
        ull p = modpow(a%n, d, n), i = s;
        while (p != 1 && p != n-1 && a % n && i--)
            p = modmul(p, p, n);
        if (p != n-1 && i != s) return 0;
    }
    return 1;
}
```

Factor.h

**Description:** Pollard-rho randomized factorization algorithm. Returns prime factors of a number, in arbitrary order (e.g. 2299 -> {11, 19, 11}).

**Time:**  $\mathcal{O}\left(n^{1/4}\right)$ , less for numbers with small factors.  
"ModMulLL.h", "MillerRabin.h" d8d98d, 18 lines

```
ull pollard(ull n) {
```

```
    ull x = 0, y = 0, t = 30, prd = 2, i = 1, q;
    auto f = [&](ull x) { return modmul(x, x, n) + i; };
    while (t++ % 40 || __gcd(prd, n) == 1) {
        if (x == y) x = ++i, y = f(x);
        if ((q = modmul(prd, max(x,y) - min(x,y), n)) prd = q;
            x = f(x), y = f(f(y));
        }
        return __gcd(prd, n);
    }
}
```

```
vector<ull> factor(ull n) {
    if (n == 1) return {};
    if (isPrime(n)) return {n};
    ull x = pollard(n);
    auto l = factor(x), r = factor(n / x);
    l.insert(l.end(), all(r));
    return l;
}
```

5.3 Divisibility

euclid.h

**Description:** Finds two integers  $x$  and  $y$ , such that  $ax + by = \gcd(a, b)$ . If you just need gcd, use the built in `__gcd` instead. If  $a$  and  $b$  are coprime, then  $x$  is the inverse of  $a \pmod b$ . 33ba8f, 5 lines

```
ll euclid(ll a, ll b, ll &x, ll &y) {
    if (!b) return x = 1, y = 0, a;
    ll d = euclid(b, a % b, y, x);
    return y -= a/b * x, d;
}
```

CRT.h

**Description:** Chinese Remainder Theorem.  
`crt(a, m, b, n)` computes  $x$  such that  $x \equiv a \pmod m, x \equiv b \pmod n$ . If  $|a| < m$  and  $|b| < n$ ,  $x$  will obey  $0 \leq x < \text{lcm}(m, n)$ . Assumes  $mn < 2^{62}$ .  
**Time:**  $\log(n)$

```
"euclid.h" 04d93a, 7 lines
ll crt(ll a, ll m, ll b, ll n) {
    if (n > m) swap(a, b), swap(m, n);
    ll x, y, g = euclid(m, n, x, y);
    assert((a - b) % g == 0); // else no solution
    x = (b - a) % n * x % n / g * m + a;
    return x < 0 ? x + m*n/g : x;
}
```

5.3.1 Bézout’s identity

For  $a \neq 0, b \neq 0$ , then  $d = \gcd(a, b)$  is the smallest positive integer for which there are integer solutions to

$$ax + by = d$$

If  $(x, y)$  is one solution, then all solutions are given by

$$\left(x + \frac{kb}{\gcd(a,b)}, y - \frac{ka}{\gcd(a,b)}\right), \quad k \in \mathbb{Z}$$

phiFunction.h

**Description:** Euler’s  $\phi$  function is defined as  $\phi(n) := \#$  of positive integers  $\leq n$  that are coprime with  $n$ .  $\phi(1) = 1, p$  prime  $\Rightarrow \phi(p^k) = (p-1)p^{k-1}$ ,  $m, n$  coprime  $\Rightarrow \phi(mn) = \phi(m)\phi(n)$ . If  $n = p_1^{k_1}p_2^{k_2}...p_r^{k_r}$  then  $\phi(n) = (p_1-1)p_1^{k_1-1}...(p_r-1)p_r^{k_r-1}$ .  $\phi(n) = n \cdot \prod_{p|n} (1 - 1/p)$ .  $\sum_{d|n} \phi(d) = n, \sum_{1 \leq k \leq n, \gcd(k,n)=1} k = n\phi(n)/2, n > 1$ .  
**Euler’s thm:**  $a, n$  coprime  $\Rightarrow a^{\phi(n)} \equiv 1 \pmod n$ .  
**Fermat’s little thm:**  $p$  prime  $\Rightarrow a^{p-1} \equiv 1 \pmod p \ \forall a$ . cf7d6d, 8 lines

```
const int LIM = 5000000;
int phi[LIM];

void calculatePhi() {
    rep(i, 0, LIM) phi[i] = i&1 ? i : i/2;
    for (int i = 3; i < LIM; i += 2) if(phi[i] == i)
        for (int j = i; j < LIM; j += i) phi[j] -= phi[j] / i;
}
```

5.4 Fractions

FracBinarySearch.h

**Description:** Given  $f$  and  $N$ , finds the smallest fraction  $p/q \in [0, 1]$  such that  $f(p/q)$  is true, and  $p, q \leq N$ . You may want to throw an exception from  $f$  if it finds an exact solution, in which case  $N$  can be removed.

**Usage:** `fracBS([](Frac f) { return f.p>=3*f.q; }, 10);` // {1,3}  
**Time:**  $\mathcal{O}(\log(N))$  27ab3e, 25 lines

```
struct Frac { ll p, q; };
```

```
template<class F>
Frac fracBS(F f, ll N) {
    bool dir = 1, A = 1, B = 1;
    Frac lo{0, 1}, hi{1, 1}; // Set hi to 1/0 to search (0, N]
    if (f(lo)) return lo;
    assert(f(hi));
    while (A || B) {
        ll adv = 0, step = 1; // move hi if dir, else lo
        for (int si = 0; step; (step *= 2) >= si) {
            adv += step;
            Frac mid{lo.p * adv + hi.p, lo.q * adv + hi.q};
            if (abs(mid.p) > N || mid.q > N || dir == !f(mid)) {
                adv -= step; si = 2;
            }
        }
        hi.p += lo.p * adv;
        hi.q += lo.q * adv;
        dir = !dir;
        swap(lo, hi);
        A = B; B = !!adv;
    }
    return dir ? hi : lo;
}
```

5.5 Pythagorean Triples

The Pythagorean triples are uniquely generated by

$$a = k \cdot (m^2 - n^2), \quad b = k \cdot (2mn), \quad c = k \cdot (m^2 + n^2),$$

with  $m > n > 0, k > 0, m \perp n$ , and either  $m$  or  $n$  even.

5.6 Primes

$p = 962592769$  is such that  $2^{21} \mid p-1$ , which may be useful. For hashing use 970592641 (31-bit number), 31443539979727 (45-bit), 3006703054056749 (52-bit). There are 78498 primes less than 1 000 000.

Primitive roots exist modulo any prime power  $p^a$ , except for  $p = 2, a > 2$ , and there are  $\phi(\phi(p^a))$  many. For  $p = 2, a > 2$ , the group  $\mathbb{Z}_{2^a}^\times$  is instead isomorphic to  $\mathbb{Z}_2 \times \mathbb{Z}_{2^{a-2}}$ .

5.7 Estimates

$$\sum_{d|n} d = \mathcal{O}(n \log \log n).$$

The number of divisors of  $n$  is at most around 100 for  $n < 5e4$ , 500 for  $n < 1e7$ , 2000 for  $n < 1e10$ , 200 000 for  $n < 1e19$ .

5.8 Mobius Function

$$\mu(n) = \begin{cases} 0 & n \text{ is not square free} \\ 1 & n \text{ has even number of prime factors} \\ -1 & n \text{ has odd number of prime factors} \end{cases}$$

Mobius Inversion:

$$g(n) = \sum_{d|n} f(d) \Leftrightarrow f(n) = \sum_{d|n} \mu(d)g(n/d)$$

Other useful formulas/forms:

$$\sum_{d|n} \mu(d) = [n = 1] \text{ (very useful)}$$

$$g(n) = \sum_{n|d} f(d) \Leftrightarrow f(n) = \sum_{n|d} \mu(d/n)g(d)$$

$$g(n) = \sum_{1 \leq m \leq n} f(\lfloor \frac{n}{m} \rfloor) \Leftrightarrow f(n) = \sum_{1 \leq m \leq n} \mu(m)g(\lfloor \frac{n}{m} \rfloor)$$

Graph (6)

6.1 Fundamentals

BellmanFord.h

**Description:** Calculates shortest paths from  $s$  in a graph that might have negative edge weights. Unreachable nodes get  $\text{dist} = \text{inf}$ ; nodes reachable through negative-weight cycles get  $\text{dist} = -\text{inf}$ . Assumes  $V^2 \max |w_i| < \sim 2^{83}$ .  
**Time:**  $\mathcal{O}(VE)$

830a8f, 23 lines

```
const ll inf = LLONG_MAX;
struct Ed { int a, b, w, s() { return a < b ? a : -a; }};
struct Node { ll dist = inf; int prev = -1; };
```

```
void bellmanFord(vector<Node>& nodes, vector<Ed>& eds, int s) {
    nodes[s].dist = 0;
    sort(all(eds), [](Ed a, Ed b) { return a.s() < b.s(); });

    int lim = sz(nodes) / 2 + 2; // /3+100 with shuffled vertices
    rep(i,0,lim) for (Ed ed : eds) {
        Node cur = nodes[ed.a], &dest = nodes[ed.b];
        if (abs(cur.dist) == inf) continue;
        ll d = cur.dist + ed.w;
        if (d < dest.dist) {
            dest.prev = ed.a;
            dest.dist = (i < lim-1 ? d : -inf);
        }
    }
    rep(i,0,lim) for (Ed e : eds) {
        if (nodes[e.a].dist == -inf)
            nodes[e.b].dist = -inf;
    }
}
```

FloydWarshall.h

**Description:** Calculates all-pairs shortest path in a directed graph that might have negative edge weights. Input is an distance matrix  $m$ , where  $m[i][j] = \text{inf}$  if  $i$  and  $j$  are not adjacent. As output,  $m[i][j]$  is set to the shortest distance between  $i$  and  $j$ ,  $\text{inf}$  if no path, or  $-\text{inf}$  if the path goes through a negative-weight cycle.  
**Time:**  $\mathcal{O}(N^3)$

531245, 12 lines

```
const ll inf = 1LL << 62;
```

```
void floydWarshall(vector<vector<ll>>& m) {
    int n = sz(m);
    rep(i,0,n) m[i][i] = min(m[i][i], 0LL);
    rep(k,0,n) rep(i,0,n) rep(j,0,n)
        if (m[i][k] != inf && m[k][j] != inf) {
            auto newDist = max(m[i][k] + m[k][j], -inf);
            m[i][j] = min(m[i][j], newDist);
        }
    rep(k,0,n) if (m[k][k] < 0) rep(i,0,n) rep(j,0,n)
        if (m[i][k] != inf && m[k][j] != inf) m[i][j] = -inf;
}
```

6.2 Network flow

MinCostMaxFlow.h

**Description:** Min-cost max-flow. If costs can be negative, call `setpi` before `maxflow`, but note that negative cost cycles are not supported. To obtain the actual flow, look at positive values only.  
**Time:**  $\mathcal{O}(FE \log(V))$  where  $F$  is max flow.  $\mathcal{O}(VE)$  for `setpi`.

58385b, 79 lines

```
#include <bits/extc++.h>

const ll INF = numeric_limits<ll>::max() / 4;

struct MCMF {
    struct edge {
        int from, to, rev;
        ll cap, cost, flow;
    };
    int N;
    vector<vector<edge>> ed;
    vi seen;
    vector<ll> dist, pi;
    vector<edge*> par;

    MCMF(int N) : N(N), ed(N), seen(N), dist(N), pi(N), par(N) {}

    void addEdge(int from, int to, ll cap, ll cost) {
        if (from == to) return;
        ed[from].push_back(edge{ from,to,sz(ed[to]),cap,cost,0 });
        ed[to].push_back(edge{ to,from,sz(ed[from])-1,0,-cost,0 });
    }

    void path(int s) {
        fill(all(seen), 0);
        fill(all(dist), INF);
        dist[s] = 0; ll di;

        __gnu_pbds::priority_queue<pair<ll, int>> q;
        vector<decltype(q)::point_iterator> its(N);
        q.push({ 0, s });

        while (!q.empty()) {
            s = q.top().second; q.pop();
            seen[s] = 1; di = dist[s] + pi[s];
            for (edge& e : ed[s]) if (!seen[e.to]) {
                ll val = di - pi[e.to] + e.cost;
                if (e.cap - e.flow > 0 && val < dist[e.to]) {
                    dist[e.to] = val;
                    par[e.to] = &e;
                    if (its[e.to] == q.end())
                        its[e.to] = q.push({ -dist[e.to], e.to });
                    else
                        q.modify(its[e.to], { -dist[e.to], e.to });
                }
            }
        }
        rep(i,0,N) pi[i] = min(pi[i] + dist[i], INF);
    }
}
```

```
pair<ll, ll> maxflow(int s, int t) {
    ll totflow = 0, totcost = 0;
    while (path(s), seen[t]) {
        ll fl = INF;
        for (edge* x = par[t]; x; x = par[x->from])
            fl = min(fl, x->cap - x->flow);

        totflow += fl;
        for (edge* x = par[t]; x; x = par[x->from]) {
            x->flow += fl;
            ed[x->to][x->rev].flow -= fl;
        }
    }
    rep(i,0,N) for(edge& e : ed[i]) totcost += e.cost * e.flow;
    return {totflow, totcost/2};
}
```

```
// If some costs can be negative, call this before maxflow:
void setpi(int s) { // (otherwise, leave this out)
    fill(all(pi), INF); pi[s] = 0;
    int it = N, ch = 1; ll v;
    while (ch-- && it--)
        rep(i,0,N) if (pi[i] != INF)
            for (edge& e : ed[i]) if (e.cap)
                if ((v = pi[i] + e.cost) < pi[e.to])
                    pi[e.to] = v, ch = 1;
    assert(it >= 0); // negative cost cycle
}
```

Dinic.h

**Description:** Flow algorithm with complexity  $\mathcal{O}(VE \log U)$  where  $U = \max |\text{cap}|$ .  $\mathcal{O}(\min(E^{1/2}, V^{2/3})E)$  if  $U = 1$ ;  $\mathcal{O}(\sqrt{VE})$  for bipartite matching.

d7f0f1, 42 lines

```
struct Dinic {
    struct Edge {
        int to, rev;
        ll c, oc;
        ll flow() { return max(oc - c, 0LL); } // if you need flows
    };
    vi lvl, ptr, q;
    vector<vector<Edge>> adj;
    Dinic(int n) : lvl(n), ptr(n), q(n), adj(n) {}
    void addEdge(int a, int b, ll c, ll rcap = 0) {
        adj[a].push_back({b, sz(adj[b]), c, c});
        adj[b].push_back({a, sz(adj[a]) - 1, rcap, rcap});
    }
    ll dfs(int v, int t, ll f) {
        if (v == t || !f) return f;
        for (int& i = ptr[v]; i < sz(adj[v]); i++) {
            Edge& e = adj[v][i];
            if (lvl[e.to] == lvl[v] + 1)
                if (ll p = dfs(e.to, t, min(f, e.c))) {
                    e.c -= p, adj[e.to][e.rev].c += p;
                    return p;
                }
        }
        return 0;
    }
    ll calc(int s, int t) {
        ll flow = 0; q[0] = s;
        rep(L,0,31) do { // 'int L=30' maybe faster for random data
            lvl = ptr = vi(sz(q));
            int qi = 0, qe = lvl[s] = 1;
            while (qi < qe && !lvl[t]) {
                int v = q[qi++];
                for (Edge e : adj[v])
                    if (!lvl[e.to] && e.c >> (30 - L))
```

```
        q[qe++] = e.to, lvl[e.to] = lvl[v] + 1;
    }
    while (ll p = dfs(s, t, LLONG_MAX)) flow += p;
} while (lvl[t]);
return flow;
}
bool leftOfMinCut(int a) { return lvl[a] != 0; }
};
```

GlobalMinCut.h  
**Description:** Find a global minimum cut in an undirected graph, as represented by an adjacency matrix.  
**Time:**  $\mathcal{O}(V^3)$

```
pair<int, vi> globalMinCut(vector<vi> mat) {
    pair<int, vi> best = {INT_MAX, {}};
    int n = sz(mat);
    vector<vi> co(n);
    rep(i,0,n) co[i] = {i};
    rep(ph,1,n) {
        vi w = mat[0];
        size_t s = 0, t = 0;
        rep(it,0,n-ph) { // O(V^2) -> O(E log V) with prio. queue
            w[t] = INT_MIN;
            s = t, t = max_element(all(w)) - w.begin();
            rep(i,0,n) w[i] += mat[t][i];
        }
        best = min(best, {w[t] - mat[t][t], co[t]});
        co[s].insert(co[s].end(), all(co[t]));
        rep(i,0,n) mat[s][i] += mat[t][i];
        rep(i,0,n) mat[i][s] = mat[s][i];
        mat[0][t] = INT_MIN;
    }
    return best;
}
```

### 6.3 Matching

#### hopcroftKarp.h

**Description:** Fast bipartite matching algorithm. Graph  $g$  should be a list of neighbors of the left partition, and  $btoa$  should be a vector full of -1's of the same size as the right partition. Returns the size of the matching.  $btoa[i]$  will be the match for vertex  $i$  on the right side, or  $-1$  if it's not matched.  
**Usage:** vi btoa(m, -1); hopcroftKarp(g, btoa);  
**Time:**  $\mathcal{O}(\sqrt{VE})$

```
bool dfs(int a, int L, vector<vi>& g, vi& btoa, vi& A, vi& B) {
    if (A[a] != L) return 0;
    A[a] = -1;
    for (int b : g[a]) if (B[b] == L + 1) {
        B[b] = 0;
        if (btoa[b] == -1 || dfs(btoa[b], L + 1, g, btoa, A, B))
            return btoa[b] = a, 1;
    }
    return 0;
}

int hopcroftKarp(vector<vi>& g, vi& btoa) {
    int res = 0;
    vi A(g.size()), B(btoa.size()), cur, next;
    for (;;) {
        fill(all(A), 0);
        fill(all(B), 0);
        cur.clear();
        for (int a : btoa) if (a != -1) A[a] = -1;
        rep(a,0,sz(g)) if (A[a] == 0) cur.push_back(a);
        for (int lay = 1; ; lay++) {
            bool islast = 0;
            next.clear();
```

```
        for (int a : cur) for (int b : g[a]) {
            if (btoa[b] == -1) {
                B[b] = lay;
                islast = 1;
            }
            else if (btoa[b] != a && !B[b]) {
                B[b] = lay;
                next.push_back(btoa[b]);
            }
        }
        if (islast) break;
        if (next.empty()) return res;
        for (int a : next) A[a] = lay;
        cur.swap(next);
    }
    rep(a,0,sz(g))
        res += dfs(a, 0, g, btoa, A, B);
}

}
```

#### DFSMatching.h

**Description:** Simple bipartite matching algorithm. Graph  $g$  should be a list of neighbors of the left partition, and  $btoa$  should be a vector full of -1's of the same size as the right partition. Returns the size of the matching.  $btoa[i]$  will be the match for vertex  $i$  on the right side, or  $-1$  if it's not matched.  
**Usage:** vi btoa(m, -1); dfsMatching(g, btoa);  
**Time:**  $\mathcal{O}(VE)$

```
bool find(int j, vector<vi>& g, vi& btoa, vi& vis) {
    if (btoa[j] == -1) return 1;
    vis[j] = 1; int di = btoa[j];
    for (int e : g[di])
        if (!vis[e] && find(e, g, btoa, vis)) {
            btoa[e] = di;
            return 1;
        }
    return 0;
}

int dfsMatching(vector<vi>& g, vi& btoa) {
    vi vis;
    rep(i,0,sz(g)) {
        vis.assign(sz(btoa), 0);
        for (int j : g[i])
            if (find(j, g, btoa, vis)) {
                btoa[j] = i;
                break;
            }
    }
    return sz(btoa) - (int)count(all(btoa), -1);
}
```

#### MinimumVertexCover.h

**Description:** Finds a minimum vertex cover in a bipartite graph. The size is the same as the size of a maximum matching, and the complement is a maximum independent set.

```
"DFSMatching.h"
vi cover(vector<vi>& g, int n, int m) {
    vi match(m, -1);
    int res = dfsMatching(g, match);
    vector<bool> lfound(n, true), seen(m);
    for (int it : match) if (it != -1) lfound[it] = false;
    vi q, cover;
    rep(i,0,n) if (lfound[i]) q.push_back(i);
    while (!q.empty()) {
        int i = q.back(); q.pop_back();
        lfound[i] = 1;
        for (int e : g[i]) if (!seen[e] && match[e] != -1) {
            seen[e] = true;
            q.push_back(match[e]);
        }
    }
```

```
    }
}
rep(i,0,n) if (!lfound[i]) cover.push_back(i);
rep(i,0,m) if (seen[i]) cover.push_back(n+i);
assert(sz(cover) == res);
return cover;
}
```

#### WeightedMatching.h

**Description:** Given a weighted bipartite graph, matches every node on the left with a node on the right such that no nodes are in two matchings and the sum of the edge weights is minimal. Takes cost[N][M], where cost[i][j] = cost for L[i] to be matched with R[j] and returns (min cost, match), where L[i] is matched with R[match[i]]. Negate costs for max cost. Requires  $N \leq M$ .  
**Time:**  $\mathcal{O}(N^2M)$

```
pair<int, vi> hungarian(const vector<vi> &a) {
    if (a.empty()) return {0, {}};
    int n = sz(a) + 1, m = sz(a[0]) + 1;
    vi u(n), v(m), p(m), ans(n - 1);
    rep(i,1,n) {
        p[0] = i;
        int j0 = 0; // add "dummy" worker 0
        vi dist(m, INT_MAX), pre(m, -1);
        vector<bool> done(m + 1);
        do { // dijkstra
            done[j0] = true;
            int i0 = p[j0], j1, delta = INT_MAX;
            rep(j,1,m) if (!done[j]) {
                auto cur = a[i0 - 1][j - 1] - u[i0] - v[j];
                if (cur < dist[j]) dist[j] = cur, pre[j] = j0;
                if (dist[j] < delta) delta = dist[j], j1 = j;
            }
            rep(j,0,m) {
                if (done[j]) u[p[j]] += delta, v[j] -= delta;
                else dist[j] -= delta;
            }
            j0 = j1;
        } while (p[j0]);
        while (j0) { // update alternating path
            int j1 = pre[j0];
            p[j0] = p[j1], j0 = j1;
        }
    }
    rep(j,1,m) if (p[j]) ans[p[j] - 1] = j - 1;
    return {-v[0], ans}; // min cost
}
```

#### GeneralMatching.h

**Description:** Matching for general graphs. If  $i$  is not matched, match[i] =  $n$   
**Usage:** G.add\_edge(u,v), G.solve()  
**Time:**  $\mathcal{O}(N^3)$

```
struct Matching { // 0-base
    queue<int> q; int n;
    v<int> fa, s, vis, pre, match;
    vv<int> G;
    int Find(int u)
    { return u == fa[u] ? u : fa[u] = Find(fa[u]); }
    int LCA(int x, int y) {
        static int tk = 0; tk++; x = Find(x); y = Find(y);
        for (;;) swap(x, y) if (x != n) {
            if (vis[x] == tk) return x;
            vis[x] = tk;
            x = Find(pre[match[x]]);
        }
    }
    void Blossom(int x, int y, int l) {
```

```

    for (; Find(x) != 1; x = pre[y]) {
        pre[x] = y, y = match[x];
        if (s[y] == 1) q.push(y), s[y] = 0;
        for (int z: {x, y}) if (fa[z] == z) fa[z] = 1;
    }
}
bool Bfs(int r) {
    iota(all(fa), 0); fill(all(s), -1);
    q = queue<int>(); q.push(r); s[r] = 0;
    for (; !q.empty(); q.pop()) {
        for (int x = q.front(); int u : G[x])
            if (s[u] == -1) {
                if (pre[u] = x, s[u] = 1, match[u] == n) {
                    for (int a = u, b = x, last;
                        b != n; a = last, b = pre[a])
                        last = match[b], match[b] = a, match[a] = b;
                    return true;
                }
                q.push(match[u]); s[match[u]] = 0;
            } else if (!s[u] && Find(u) != Find(x)) {
                int l = LCA(u, x);
                Blossom(x, u, l); Blossom(u, x, l);
            }
        }
    return false;
}
Matching(int _n) : n(_n), fa(n + 1), s(n + 1), vis(n + 1),
    pre(n + 1, n), match(n + 1, n), G(n) {}
void add_edge(int u, int v)
{ G[u].pb(v), G[v].pb(u); }
int solve() {
    int ans = 0;
    for (int x = 0; x < n; ++x)
        if (match[x] == n) ans += Bfs(x);
    return ans;
} // match[x] == n means not matched
};

```

## 6.4 DFS algorithms

### 2CC.h

**Description:** All 2CC are returned in ans

3e9fe4, 29 lines

```

const int N = 2e5;
vector<int>adj[N];
int tin[N], low[N], timer = 0;
vector<vector<int>>ans;
stack<int>stk;
void dfs(int v, int u) {
    bool p = 0;
    tin[v] = low[v] = ++timer;
    stk.push(v);
    for (int x: adj[v]) {
        if (x != u || p) {
            if (!tin[x]) {
                dfs(x, v);
                low[v] = min(low[v], low[x]);
            } else low[v] = min(low[v], tin[x]);
        }
        else p = 1;
    }
    if (low[v] == tin[v]) {
        vector<int>c;
        while (stk.top() != v) {
            c.pb(stk.top());
            stk.pop();
        }
        c.pb(v);
        stk.pop();
        ans.pb(c);
    }
}

```

```

    }
}
}
BCC.h
Description: Returns the block cut tree in tree Block nodes labels >= n A
node is an articulation point if it is connected to exactly one block node
26f25a, 32 lines
struct BCC {
    int n, n0, timer = 0;
    stack<int>stk;
    v<int>tin, low;
    vv<int>tree;
    void dfs(int v, int p, vv<pii>&g) {
        tin[v] = low[v] = ++timer;
        stk.push(v);
        for (auto [x, id]: g[v]) {
            if (id == p) continue;
            if (tin[x]) low[v] = min(low[v], tin[x]);
            else {
                dfs(x, id, g);
                low[v] = min(low[v], low[x]);
                if (low[x] >= tin[v]) {
                    tree[v].pb(n0);
                    tree[n0].pb(v);
                    while (tree[n0].back() != x) {
                        tree[n0].pb(stk.top());
                        tree[stk.top()].pb(n0);
                        stk.pop();
                    }
                    n0++;
                }
            }
        }
    }
    BCC(int _n): n(_n), n0(_n), tin(_n), low(_n), tree(2*_n) {}
    void build(vv<pii>&g) {
        for (int i=0; i<n; i++) if (!tin[i]) dfs(i, -1, g);
    }
};

```

### SCC.h

**Description:** comp[i] holds SCC index of i, note that SCC index is sorted in reverse topological order

**Time:**  $\mathcal{O}(E + V)$

e756da, 27 lines

```

struct SCC {
    int n, timer = 0, ccnt = 0;
    vector<int>low, tin, comp;
    stack<int>stk;
    SCC(int _n): n(_n), low(n, 1e9), tin(n), comp(n) {}
    void dfs(int v, vv<int>&g) {
        stk.push(v);
        low[v] = tin[v] = ++timer;
        for (int x: g[v]) {
            if (!tin[x]) dfs(x, g);
            low[v] = min(low[v], low[x]);
        }
        if (tin[v] == low[v]) {
            while (true) {
                int j = stk.top();
                stk.pop();
                comp[j] = ccnt;
                low[j] = 1e9;
                if (j == v) break;
            }
            ccnt++;
        }
    }
    void build(vv<int> g) {

```

```

        for (int i=0; i<n; i++) if (!tin[i]) dfs(i, g);
    }
};

```

### 2sat.h

**Description:** Calculates a valid assignment to boolean variables a, b, c, ... to a 2-SAT problem, so that an expression of the type  $(a|b) \&\& (!a|c) \&\& (d|!b) \&\& \dots$  becomes true, or reports that it is unsatisfiable. Negated variables are represented by bit-inversions ( $\sim x$ ).

**Usage:** TwoSat ts(number of boolean variables);  
ts.either(0, ~3); // Var 0 is true or var 3 is false  
ts.setValue(2); // Var 2 is true  
ts.atMostOne({0, ~1, 2}); //  $\leq 1$  of vars 0, ~1 and 2 are true  
ts.solve(); // Returns true iff it is solvable  
ts.values[0..N-1] holds the assigned values to the vars

**Time:**  $\mathcal{O}(N + E)$ , where N is the number of boolean variables, and E is the number of clauses.

5f9706, 56 lines

```

struct TwoSat {
    int N;
    vector<vi> gr;
    vi values; // 0 = false, 1 = true

    TwoSat(int n = 0) : N(n), gr(2*n) {}

    int addVar() { // (optional)
        gr.emplace_back();
        gr.emplace_back();
        return N++;
    }

    void either(int f, int j) {
        f = max(2*f, -1-2*f);
        j = max(2*j, -1-2*j);
        gr[f].push_back(j^1);
        gr[j].push_back(f^1);
    }

    void setValue(int x) { either(x, x); }

    void atMostOne(const vi& li) { // (optional)
        if (sz(li) <= 1) return;
        int cur = ~li[0];
        rep(i, 2, sz(li)) {
            int next = addVar();
            either(cur, ~li[i]);
            either(cur, next);
            either(~li[i], next);
            cur = ~next;
        }
        either(cur, ~li[1]);
    }

    vi val, comp, z; int time = 0;
    int dfs(int i) {
        int low = val[i] = ++time, x; z.push_back(i);
        for (int e : gr[i]) if (!comp[e])
            low = min(low, val[e] ? dfs(e));
        if (low == val[i]) do {
            x = z.back(); z.pop_back();
            comp[x] = low;
            if (values[x>>1] == -1)
                values[x>>1] = x&1;
        } while (x != i);
        return val[i] = low;
    }

    bool solve() {
        values.assign(N, -1);
        val.assign(2*N, 0); comp = val;

```

```
    rep(i,0,2*N) if (!comp[i]) dfs(i);
    rep(i,0,N) if (comp[2*i] == comp[2*i+1]) return 0;
    return 1;
}
};
```

EulerWalk.h  
**Description:** Eulerian undirected/directed path/cycle algorithm. Input is adjacency list of edge,edge index Undirected, two copies (u,v),(v,u) with same index for euler paths, set path = 1 to return edge index, make s contains pairs  
**Time:**  $\mathcal{O}(V + E)$

fe22e0, 15 lines

```
vi eulerWalk(vector<vector<pii>>& gr, int nedges, int src,bool
    path) {
    int n = sz(gr);
    vi D(n), its(n), eu(nedges), ret, s = {src};
    if(path)D[src]++; // to allow Euler paths, not just cycles
    while (!s.empty()) {
        int x = s.back(), y, e, &it = its[x], end = sz(gr[x]);
        if (it == end){ ret.push_back(x); s.pop_back(); continue; }
        tie(y, e) = gr[x][it++];
        if (!eu[e]) {
            D[x]--, D[y]++;
            eu[e] = 1; s.push_back(y);
        }
    }
    for (int x : D) if (x < 0 || sz(ret) != nedges+1) return {};
    return {ret.rbegin(), ret.rend()};
}
```

DominatorTree.h  
**Description:** Find the immediate dominators for all nodes  
**Usage:** domtree d(n),d.build(source,adj);  
**Time:**  $\mathcal{O}(E + V)$

fa59e4, 55 lines

```
struct dom_tree{
    int n,timer = 0;
    v<int>tin,sdom,idom,a,p,pmin,ord;
    vv<int>g,cross,R;
    dom_tree(int _n : n(_n),tin(n),sdom(n),idom(n,-1),
        a(n),p(n),pmin(n),g(n),cross(n),R(n) {});
    inline int Min0(int a,int b){
        return tin[a] < tin[b]? a : b;
    }
    inline int Min1(int a,int b){
        return tin[sdom[a]] < tin[sdom[b]]? a : b;
    }

    int find(int x,bool mode){
        if(a[x] == -1)return x;
        if(!mode)pmin[x] = Min0(pmin[x],find(a[x],0));
        else pmin[x] = Min1(pmin[x],find(a[x],1));
        if(a[a[x]] != -1)a[x] = a[a[x]];
        return pmin[x];
    }

    void dfs1(int v,vv<int>&adj){
        ord.push_back(v);
        tin[v] = ++timer;
        for(int x:adj[v]){
            cross[x].push_back(v);
            if(tin[x])continue;
            dfs1(x,adj);
            p[x] = v;
            g[v].push_back(x);
        }
    }

    void build(int s,vv<int>&adj){
        dfs1(s,adj);
        a.assign(n,-1);
        reverse(all(ord));
```

```
    for(int v:ord){
        sdom[v] = v;
        if(v == s)break;
        for(int u:cross[v])
            sdom[v] = Min0(sdom[v],find(u,0));
        a[v] = p[v];
        pmin[v] = sdom[v];
    }
    a.assign(n,-1);
    iota(all(pmin),0);
    for(int v:ord){
        for(int u:R[v])idom[u] = find(u,1);
        if(v == s){idom[s] = s;break;}
        for(int u:g[v])a[u] = p[u];
        R[sdom[v]].push_back(v);
    }
    reverse(all(ord));
    for(int v:ord)
        idom[v] = idom[v] == v?sdom[v]:idom[idom[v]];
}
};
```

6.5 Coloring  
EdgeColoring.h  
**Description:** Given a simple, undirected graph with max degree  $D$ , computes a  $(D + 1)$ -coloring of the edges such that no neighboring edges share a color. ( $D$ -coloring is NP-hard, but can be done for bipartite graphs by repeated matchings of max-degree nodes.)  
**Time:**  $\mathcal{O}(NM)$

e210e2, 31 lines

```
vi edgeColoring(int N, vector<pii> eds) {
    vi cc(N + 1), ret(sz(eds)), fan(N), free(N), loc;
    for (pii e : eds) ++cc[e.first], ++cc[e.second];
    int u, v, ncols = *max_element(all(cc)) + 1;
    vector<vi> adj(N, vi(ncols, -1));
    for (pii e : eds) {
        tie(u, v) = e;
        fan[0] = v;
        loc.assign(ncols, 0);
        int at = u, end = u, d, c = free[u], ind = 0, i = 0;
        while (d = free[v], !loc[d] && (v = adj[u][d]) != -1)
            loc[d] = ++ind, cc[ind] = d, fan[ind] = v;
        cc[loc[d]] = c;
        for (int cd = d; at != -1; cd ^= c ^ d, at = adj[at][cd])
            swap(adj[at][cd], adj[end = at][cd ^ c ^ d]);
        while (adj[fan[i]][d] != -1) {
            int left = fan[i], right = fan[++i], e = cc[i];
            adj[u][e] = left;
            adj[left][e] = u;
            adj[right][e] = -1;
            free[right] = e;
        }
        adj[u][d] = fan[i];
        adj[fan[i]][d] = u;
        for (int y : {fan[0], u, end})
            for (int& z = free[y] = 0; adj[y][z] != -1; z++);
    }
    rep(i,0,sz(eds))
        for (tie(u, v) = eds[i]; adj[u][ret[i]] != v;) ++ret[i];
    return ret;
}
```

6.6 Heuristics  
MaximumClique.h  
**Description:** Quickly finds a maximum clique of a graph (given as symmetric bitset matrix; self-edges not allowed). Can be used to find a maximum independent set by finding a clique of the complement graph.

**Time:** Runs in about 1s for n=155 and worst case random graphs (p=.90).  
Runs faster for sparse graphs.

f7c0bc, 49 lines

```
typedef vector<bitset<200>> vb;
struct MaxClique {
    double limit=0.025, pk=0;
    struct Vertex { int i, d=0; };
    typedef vector<Vertex> vv;
    vb e;
    vv V;
    vector<vi> C;
    vi qmax, q, S, old;
    void init(vv& r) {
        for (auto& v : r) v.d = 0;
        for (auto& v : r) for (auto j : r) v.d += e[v.i][j.i];
        sort(all(r), [](auto a, auto b) { return a.d > b.d; });
        int mxD = r[0].d;
        rep(i,0,sz(r)) r[i].d = min(i, mxD) + 1;
    }
    void expand(vv& R, int lev = 1) {
        S[lev] += S[lev - 1] - old[lev];
        old[lev] = S[lev - 1];
        while (sz(R)) {
            if (sz(q) + R.back().d <= sz(qmax)) return;
            q.push_back(R.back().i);
            vv T;
            for(auto v:R) if (e[R.back().i][v.i]) T.push_back({v.i});
            if (sz(T)) {
                if (S[lev]++ / ++pk < limit) init(T);
                int j = 0, mxk = 1, mnk = max(sz(qmax) - sz(q) + 1, 1);
                C[1].clear(), C[2].clear();
                for (auto v : T) {
                    int k = 1;
                    auto f = [&](int i) { return e[v.i][i]; };
                    while (any_of(all(C[k]), f)) k++;
                    if (k > mxk) mxk = k, C[mxk + 1].clear();
                    if (k < mnk) T[j++].i = v.i;
                    C[k].push_back(v.i);
                }
                if (j > 0) T[j - 1].d = 0;
                rep(k,mnk,mxk + 1) for (int i : C[k])
                    T[j].i = i, T[j++].d = k;
                expand(T, lev + 1);
            } else if (sz(q) > sz(qmax)) qmax = q;
            q.pop_back(), R.pop_back();
        }
    }
    vi maxClique() { init(V), expand(V); return qmax; }
    MaxClique(vb conn) : e(conn), C(sz(e)+1), S(sz(C)), old(S) {
        rep(i,0,sz(e)) V.push_back({i});
    }
};
```

MaximumIndependentSet.h  
**Description:** To obtain a maximum independent set of a graph, find a max clique of the complement. If the graph is bipartite, see MinimumVertex-Cover.

6.7 Trees  
BinaryLifting.h  
**Description:** Calculate power of two jumps in a tree, to support fast upward jumps and LCAs. Assumes the root node points to itself.  
**Time:** construction  $\mathcal{O}(N \log N)$ , queries  $\mathcal{O}(\log N)$

bfce85, 25 lines

```
vector<vi> treeJump(vi& P){
    int on = 1, d = 1;
    while (on < sz(P)) on *= 2, d++;
    vector<vi> jmp(d, P);
```

```

    rep(i,1,d) rep(j,0,sz(P))
        jmp[i][j] = jmp[i-1][jmp[i-1][j]];
    return jmp;
}

int jmp(vector<vi>& tbl, int nod, int steps){
    rep(i,0,sz(tbl))
        if(steps&(1<<i)) nod = tbl[i][nod];
    return nod;
}

```

```

int lca(vector<vi>& tbl, vi& depth, int a, int b) {
    if (depth[a] < depth[b]) swap(a, b);
    a = jmp(tbl, a, depth[a] - depth[b]);
    if (a == b) return a;
    for (int i = sz(tbl); i--;) {
        int c = tbl[i][a], d = tbl[i][b];
        if (c != d) a = c, b = d;
    }
    return tbl[0][a];
}

```

## LCA.h

**Description:** Data structure for computing lowest common ancestors in a tree (with 0 as root). C should be an adjacency list of the tree, either directed or undirected.

**Time:**  $\mathcal{O}(N \log N + Q)$

../data-structures/RMQ.h 0f62fb, 21 lines

```

struct LCA {
    int T = 0;
    vi time, path, ret;
    RMQ<int> rmq;

    LCA(vector<vi>& C) : time(sz(C)), rmq((dfs(C,0,-1), ret)) {}
    void dfs(vector<vi>& C, int v, int par) {
        time[v] = T++;
        for (int y : C[v]) if (y != par) {
            path.push_back(v), ret.push_back(time[v]);
            dfs(C, y, v);
        }
    }

    int lca(int a, int b) {
        if (a == b) return a;
        tie(a, b) = minmax(time[a], time[b]);
        return path[rmq.query(a, b)];
    }
    //dist(a,b){return depth[a] + depth[b] - 2*depth[lca(a,b)];}
};

```

## CompressTree.h

**Description:** Given a rooted tree and a subset S of nodes, compute the minimal subtree that contains all the nodes by adding all (at most  $|S| - 1$ ) pairwise LCA's and compressing edges. Returns a list of (par, orig.index) representing a tree rooted at 0. The root points to itself.

**Time:**  $\mathcal{O}(|S| \log |S|)$

"LCA.h" 9775a0, 21 lines

```

typedef vector<pair<int, int>> vpi;
vpi compressTree(LCA& lca, const vi& subset) {
    static vi rev; rev.resize(sz(lca.time));
    vi li = subset, &T = lca.time;
    auto cmp = [&](int a, int b) { return T[a] < T[b]; };
    sort(all(li), cmp);
    int m = sz(li)-1;
    rep(i,0,m) {
        int a = li[i], b = li[i+1];
        li.push_back(lca.lca(a, b));
    }
}

```

```

sort(all(li), cmp);
li.erase(unique(all(li)), li.end());
rep(i,0,sz(li)) rev[li[i]] = i;
vpi ret = {pii(0, li[0])};
rep(i,0,sz(li)-1) {
    int a = li[i], b = li[i+1];
    ret.emplace_back(rev[lca.lca(a, b)], b);
}
return ret;
}

```

## HLD.h

**Description:** Decomposes a tree into vertex disjoint heavy paths and light edges such that the path from any leaf to the root contains at most  $\log(n)$  light edges. Code does additive modifications and max queries, but can support commutative segtree modifications/queries on paths and subtrees. Takes as input the full adjacency list. VALS.EDGES being true means that values are stored in the edges, as opposed to the nodes. All values initialized to the segtree default. Root must be 0.

**Time:**  $\mathcal{O}((\log N)^2)$

9547af, 46 lines

```

template <bool VALS_EDGES> struct HLD {
    int N, tim = 0;
    vector<vi> adj;
    vi par, siz, rt, pos;
    Node *tree;
    HLD(vector<vi> adj_)
        : N(sz(adj_)), adj(adj_), par(N, -1), siz(N, 1),
          rt(N), pos(N), tree(new Node(0, N)) { dfsSz(0); dfsHld(0); }
    void dfsSz(int v) {
        for (int& u : adj[v]) {
            adj[u].erase(find(all(adj[u]), v));
            par[u] = v;
            dfsSz(u);
            siz[v] += siz[u];
            if (siz[u] > siz[adj[v][0]]) swap(u, adj[v][0]);
        }
    }
    void dfsHld(int v) {
        pos[v] = tim++;
        for (int u : adj[v]) {
            rt[u] = (u == adj[v][0] ? rt[v] : u);
            dfsHld(u);
        }
    }
    template <class B> void process(int u, int v, B op) {
        for (; v = par[rt[v]]; ) {
            if (pos[u] > pos[v]) swap(u, v);
            if (rt[u] == rt[v]) break;
            op(pos[rt[v]], pos[v] + 1);
        }
        op(pos[u] + VALS_EDGES, pos[v] + 1);
    }
    void modifyPath(int u, int v, int val) {
        process(u, v, [&](int l, int r) { tree->add(l, r, val); });
    }
    int queryPath(int u, int v) { // Modify depending on problem
        int res = -1e9;
        process(u, v, [&](int l, int r) {
            res = max(res, tree->query(l, r));
        });
        return res;
    }
    int querySubtree(int v) { // modifySubtree is similar
        return tree->query(pos[v] + VALS_EDGES, pos[v] + siz[v]);
    }
};

```

## LinkCutTree.h

**Description:** Represents a forest of unrooted trees. Supports linking, cutting, path and subtree aggregates

**Time:** All operations take amortized  $\mathcal{O}(\log N)$ .

c6aa18, 98 lines

```

struct SplayTree {
    struct Node {
        int ch[2] = {0, 0}, p = 0;
        ll self = 0, path = 0; // Path aggregates
        ll sub = 0, vir = 0; // Subtree aggregates
        bool flip = 0; // Lazy tags
    };
    vector<Node> T;
    SplayTree(int n) : T(n + 1) {}
    void push(int x) {
        if (!x || !T[x].flip) return;
        int l = T[x].ch[0], r = T[x].ch[1];

        T[l].flip ^= 1, T[r].flip ^= 1;
        swap(T[x].ch[0], T[x].ch[1]);
        T[x].flip = 0;
    }

    void pull(int x) {
        int l = T[x].ch[0], r = T[x].ch[1]; push(l); push(r);
        T[x].path = T[l].path + T[x].self + T[r].path;
        T[x].sub = T[x].vir + T[l].sub + T[r].sub + T[x].self;
    }

    void set(int x, int d, int y) {
        T[x].ch[d] = y; T[y].p = x; pull(x);
    }

    void splay(int x) {
        auto dir = [&](int x) {
            int p = T[x].p; if (!p) return -1;
            return T[p].ch[0] == x ? 0 : T[p].ch[1] == x ? 1 : -1;
        };
        auto rotate = [&](int x) {
            int y = T[x].p, z = T[y].p, dx = dir(x), dy = dir(y);
            set(y, dx, T[x].ch[!dx]);
            set(x, !dx, y);
            if (~dy) set(z, dy, x);
            T[x].p = z;
        };
        for (push(x); ~dir(x); ) {
            int y = T[x].p, z = T[y].p;
            push(z); push(y); push(x);
            int dx = dir(x), dy = dir(y);
            if (~dy) rotate(dx != dy ? x : y);
            rotate(x);
        }
    }
};

struct LinkCut : SplayTree {
    LinkCut(int n) : SplayTree(n) {}

    int access(int x) {
        int u = x, v = 0;
        for (; u; v = u, u = T[u].p) {
            splay(u);
            int& ov = T[u].ch[1];
            T[u].vir += T[ov].sub;
            T[u].vir -= T[v].sub;
            ov = v; pull(u);
        }
        return splay(x), v;
    }
}

```

```
void reroot(int x) {
    access(x); T[x].flip ^= 1; push(x);
}

void Link(int u, int v) {
    reroot(u); access(v);
    T[v].vir += T[u].sub;
    T[u].p = v; pull(v);
}

void Cut(int u, int v) {
    reroot(u); access(v);
    T[v].ch[0] = T[u].p = 0; pull(v);
}

int LCA(int u, int v) {
    if (u == v) return u;
    access(u); int ret = access(v);
    return T[u].p ? ret : 0;
}

ll Subtree(int u, int v) {
    reroot(v); access(u); return T[u].vir + T[u].self;
}

ll Path(int u, int v) {
    reroot(u); access(v); return T[v].path;
}

void Update(int u, ll v) {
    access(u); T[u].self += v; pull(u);
}
};
```

DirectedMST.h

**Description:** Finds a minimum spanning tree/arborescence of a directed graph, given a root node. If no MST exists, returns -1.

**Time:**  $\mathcal{O}(E \log V)$

```
"/data-structures/UnionFindRollback.h" 39e620, 60 lines

struct Edge { int a, b; ll w; };
struct Node {
    Edge key;
    Node *l, *r;
    ll delta;
    void prop() {
        key.w += delta;
        if (l) l->delta += delta;
        if (r) r->delta += delta;
        delta = 0;
    }
    Edge top() { prop(); return key; }
};

Node *merge(Node *a, Node *b) {
    if (!a || !b) return a ? b;
    a->prop(), b->prop();
    if (a->key.w > b->key.w) swap(a, b);
    swap(a->l, (a->r = merge(b, a->r)));
    return a;
}

void pop(Node*& a) { a->prop(); a = merge(a->l, a->r); }

pair<ll, vi> dmst(int n, int r, vector<Edge*& g) {
    RollbackUF uf(n);
    vector<Node*> heap(n);
    for (Edge e : g) heap[e.b] = merge(heap[e.b], new Node{e});
    ll res = 0;
    vi seen(n, -1), path(n), par(n);
    seen[r] = r;
```

```
vector<Edge> Q(n), in(n, {-1,-1}), comp;
deque<tuple<int, int, vector<Edge>>> cys;
rep(s,0,n) {
    int u = s, qi = 0, w;
    while (seen[u] < 0) {
        if (!heap[u]) return {-1,{};};
        Edge e = heap[u]->top();
        heap[u]->delta -= e.w, pop(heap[u]);
        Q[qi] = e, path[qi++] = u, seen[u] = s;
        res += e.w, u = uf.find(e.a);
        if (seen[u] == s) {
            Node* cyc = 0;
            int end = qi, time = uf.time();
            do cyc = merge(cyc, heap[w = path[--qi]]);
            while (uf.join(u, w));
            u = uf.find(u), heap[u] = cyc, seen[u] = -1;
            cys.push_front({u, time, {&Q[qi], &Q[end]}});
        }
    }
    rep(i,0,qi) in[uf.find(Q[i].b)] = Q[i];
}

for (auto& [u,t,comp] : cys) { // restore sol (optional)
    uf.rollback(t);
    Edge inEdge = in[u];
    for (auto& e : comp) in[uf.find(e.b)] = e;
    in[uf.find(inEdge.b)] = inEdge;
}
rep(i,0,n) par[i] = in[i].a;
return {res, par};
}
```

6.8 Math

6.8.1 Number of Spanning Trees

Create an  $N \times N$  matrix  $\text{mat}$ , and for each edge  $a \rightarrow b \in G$ , do  $\text{mat}[a][b]--$ ,  $\text{mat}[b][b]++$  (and  $\text{mat}[b][a]--$ ,  $\text{mat}[a][a]++$  if  $G$  is undirected). Remove the  $i$ th row and column and take the determinant; this yields the number of directed spanning trees rooted at  $i$  (if  $G$  is undirected, remove any row/column).

6.8.2 Erdős–Gallai theorem

A simple graph with node degrees  $d_1 \geq \dots \geq d_n$  exists iff  $d_1 + \dots + d_n$  is even and for every  $k = 1 \dots n$ ,

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k).$$

6.8.3 Euler’s formula

For a planar graph  $G$ ,  $C$  is the number of connected components

$$V - E + F = 1 + C$$

Geometry (7)

7.1 Geometric primitives

```
Point.h 598c66, 21 lines

typedef double T;
typedef complex<T> pt;
```

```
#define x real()
#define y imag()
T dot(pt v, pt w) { return (conj(v)*w).x; } // positive when
    the angle < 90
T cross(pt v, pt w) { return (conj(v)*w).y; } // positive when
    vector w is on the left side of v(<180)
T orient(pt a, pt b, pt c) { return cross(b-a,c-a); } // return
    true when c is on left side of ab
pt perp(pt p) { return {-p.y, p.x}; }
int sgn(T d) { return (T(0) < d) - (d < T(0)); }

// Following functions only work for T double
const long double PI = acos(-1);
pt scale(pt p, double a, pt c=pt{}) { return c + (p-c) * a; }
pt rot(pt p, double a, pt c=pt{}) { return c + (p-c) * polar(1.0, a); }

// given fp = f(p), fq = f(q), return f(r)
pt linearTrans(pt p, pt q, pt r, pt fp, pt fq) { return fp + (r
    -p) * (fq-fp) / (q-p); }

// return radians of the angle between vector v and w.
// To check whether an angle is greater than 90, use the sign
// of the dot product instead.
double angle(pt v, pt w) { return acos(clamp(dot(v,w) / abs(v)
    / abs(w), -1.0, 1.0)); }
```

Line.h

```
713deb, 30 lines

struct line {
    pt v; T c; // use v=(-b, a), c to represent a line ax+by=c,
    line(pt v, T c) : v(v), c(c) {}
    line(T a, T b, T c) : v({b,-a}), c(c) {} // ax + by = c
    line(pt p, pt q) : v(q-p), c(cross(v,p)) {} // From points
        P to Q

    // These work with T = int
    T side(pt p) { return cross(v,p)-c; } // positive when p is
        on the left of v
    double dist(pt p) { return abs(side(p)) / abs(v); }
    // distance between p and line, square both sides to
        prevent the precision error
    bool cmpProj(pt p, pt q) { return dot(v,p) < dot(v,q); } //
        compare projections
    line translate(pt t) { return {v, c + cross(v,t)}; } //
        translate by a vector t

    // These require T = double
    pt proj(pt p) { return p - perp(v)*side(p)/dot(v,v); }
    pt refl(pt p) { return p - perp(v)*(T)2*side(p)/dot(v,v); }
};

// requires T = double
bool inter(line l1, line l2, pt &out) {
    T d = cross(l1.v, l2.v);
    if (d == 0) return false;
    out = (l2.v*l1.c - l1.v*l2.c) / d;
    return true;
}

line bisector(line l1, line l2, bool interior) {
    assert(cross(l1.v, l2.v) != 0); // l1 and l2 cannot be
        parallel!
    double sign = interior ? 1 : -1;
    return { l2.v/abs(l2.v) + l1.v/abs(l1.v) * sign, l2.c/abs(l
        2.v) + l1.c/abs(l1.v) * sign };
}
```

| SegmentDistance.h                                                                                                                                                                                                                                                                                                                                      | b52e6d, 9 lines |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <pre>// min distance of segment ab and point p double segPoint(pt a, pt b, pt p) {     if (a != b) {         line l(a,b);         if (l.cmpProj(a,p) &amp;&amp; l.cmpProj(p,b)) // if closest to projection             return l.dist(p); // output distance to line     }     return min(abs(p-a), abs(p-b)); // otherwise distance to A or B }</pre> |                 |

| SegmentIntersection.h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3722fa, 21 lines |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <pre>// return intersect point if intersect properly // end points of a line if parrallel // empty otherwise struct cmpX {     bool operator()(pt a, pt b) const { return a.x &lt; b.x; } }; set&lt;pt, cmpX&gt; segInter(pt a, pt b, pt c, pt d) {     set&lt;pt, cmpX&gt; s;     double oa = orient(c,d,a), ob = orient(c,d,b),            oc = orient(a,b,c), od = orient(a,b,d);     // Proper intersection exists iff opposite signs     if (oa*ob &lt; 0 &amp;&amp; oc*od &lt; 0) {         s.insert( (a*ob - b*oa) / (ob-oa) ); // replace this with {0,0} when use T = int     } else {         if ( onSegment(c,d,a) ) s.insert(a);         if ( onSegment(c,d,b) ) s.insert(b);         if ( onSegment(a,b,c) ) s.insert(c);         if ( onSegment(a,b,d) ) s.insert(d);     }     return s; }</pre> |                  |

sideOf.h

**Description:** Returns where  $p$  is as seen from  $s$  towards  $e$ .  $1/0/-1 \Leftrightarrow$  left/on line/right. If the optional argument  $eps$  is given 0 is returned if  $p$  is within distance  $eps$  from the line. P is supposed to be Point<T> where T is e.g. double or long long. It uses products in intermediate steps so watch out for overflow if using int or long long.

| <p><b>Usage:</b> bool left = sideOf(p1,p2,q)==1;</p>                                                                                                                                                                                                                                                 |                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| "Point.h"                                                                                                                                                                                                                                                                                            | 3af81c, 9 lines |
| <pre>template&lt;class P&gt; int sideOf(P s, P e, P p) { return sgn(s.cross(e, p)); }  template&lt;class P&gt; int sideOf(const P&amp; s, const P&amp; e, const P&amp; p, double eps) {     auto a = (e-s).cross(p-s);     double l = (e-s).dist()*eps;     return (a &gt; l) - (a &lt; -l); }</pre> |                 |

| OnSegment.h                                                                                                     | 111659, 3 lines |
|-----------------------------------------------------------------------------------------------------------------|-----------------|
| <pre>bool onSegment(pt a, pt b, pt p) {     return orient(a,b,p) == 0 &amp;&amp; dot(a-p, b-p) &lt;= 0; }</pre> |                 |

Angle.h

**Description:** A class for ordering angles (as represented by int points and a number of rotations around the origin). Useful for rotational sweeping. Sometimes also represents points or vectors.

**Usage:** vector<Angle> v = {w[0], w[0].t360() ...}; // sorted

```
int j = 0; rep(i,0,n) { while (v[j] < v[i].t180()) ++j; }
// sweeps j such that (j-i) represents the number of positively oriented triangles with vertices at 0 and i
```

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0f0602, 35 lines |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <pre>struct Angle {     int x, y;     int t;     Angle(int x, int y, int t=0) : x(x), y(y), t(t) {}     Angle operator-(Angle b) const { return {x-b.x, y-b.y, t}; }     int half() const {         assert(x    y);         return y &lt; 0    (y == 0 &amp;&amp; x &lt; 0);     }     Angle t90() const { return {-y, x, t + (half() &amp;&amp; x &gt;= 0)}; }     Angle t180() const { return {-x, -y, t + half()}; }     Angle t360() const { return {x, y, t + 1}; } }; bool operator&lt;(Angle a, Angle b) {     // add a.dist2() and b.dist2() to also compare distances     return make_tuple(a.t, a.half(), a.y * (ll)b.x) &lt;            make_tuple(b.t, b.half(), a.x * (ll)b.y); }  // Given two points, this calculates the smallest angle between // them, i.e., the angle that covers the defined line segment. pair&lt;Angle, Angle&gt; segmentAngles(Angle a, Angle b) {     if (b &lt; a) swap(a, b);     return (b &lt; a.t180() ?            make_pair(a, b) : make_pair(b, a.t360())); } Angle operator+(Angle a, Angle b) { // point a + vector b     Angle r(a.x + b.x, a.y + b.y, a.t);     if (a.t180() &lt; r) r.t--;     return r.t180() &lt; a ? r.t360() : r; } Angle angleDiff(Angle a, Angle b) { // angle b - angle a     int tu = b.t - a.t; a.t = b.t;     return {a.x*b.x + a.y*b.y, a.x*b.y - a.y*b.x, tu - (b &lt; a)}; }</pre> |                  |

| PolarSort.h                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4bd30e, 11 lines |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <pre>bool half(pt p, pt v) {     // return 0 when arg(p) is in [v, v+pi)     return cross(v,p) &lt; 0    (cross(v,p) == 0 &amp;&amp; dot(v,p) &lt; 0); } // return 0 when arg(p) is in [0, pi) bool half(pt p) { return p.y &lt; 0    (p.y == 0 &amp;&amp; p.x &lt; 0); } void polarSort(v&lt;pt&gt; &amp;a, pt s=pt{1, 0}) {     sort(all(a), [&amp;](pt v, pt w) {         return make_tuple(half(v, s), 0) &lt; make_tuple(half(w, s), cross(v,w));     }); }</pre> |                  |

## 7.2 Circles

| CircleCircle.h                                                                                                                                                                                                            |                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| "Point.h"                                                                                                                                                                                                                 | 427906, 18 lines |
| <pre>// T = double  int circleCircle(pt o1, double r1, pt o2, double r2, pair&lt;pt,pt&gt; &amp; out) {     pt d=o2-o1;     double d2=sq(d);     if (d2 == 0) {         assert(r1 != r2);         return 0;     } }</pre> |                  |

|                                                                                                                                                                                                                                                 |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <pre>} // concentric circles double pd = (d2 + r1*r1 - r2*r2)/2; // =  O_1P  * d double h2 = r1*r1 - pd*pd/d2; // = h^2 if (h2 &gt;= 0) {     pt p = o1 + d*pd/d2, h = perp(d)*sqrt(h2/d2);     out = {p-h, p+h}; } return 1 + sgn(h2); }</pre> |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

| CircleTangents.h                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 75abd4, 12 lines |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <pre>// r2 &lt; 0 for inner tangent // r2 = 0 to find tangent line through point int tangents(pt o1, double r1, pt o2, double r2, vector&lt;pair&lt;pt ,pt&gt;&gt; &amp;out) {     pt d = o2-o1;     double dr = r1-r2, d2 = dot(d,d), h2 = d2-dr*dr;     if (d2 == 0    h2 &lt; 0) {assert(h2 != 0); return 0;}     for (double sign : {-1,1}) {         pt v = (d*dr+perp(d)*sqrt(h2)*sign)/d2;         out.push_back({o1 + v*r1, o2 + v*r2});     }     return 1 + (h2 &gt; 0); }</pre> |                  |

| CircleLine.h                                                                                                                                                                                                                                                                                                                                                                                               | f6786c, 10 lines |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <pre>// return the number of intersections int circleline(pt o, double r, line l, pair&lt;pt,pt&gt; &amp;out) {     double h2 = r*r - l.dist(o)*l.dist(o);     if (h2 &gt;= 0) { // the line touches the circle         pt p = l.proj(o); // point P         pt h = l.v*sqrt(h2)/abs(l.v); // vector parallel to l,         // of length h         out = {p-h, p+h};     }     return 1 + sgn(h2); }</pre> |                  |

| MinimumEnclosingCircle.h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <p><b>Description:</b> Computes the minimum circle that encloses a set of points.</p> <p><b>Time:</b> expected <math>\mathcal{O}(n)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                            |                  |
| "circumcircle.h"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 09dd0a, 17 lines |
| <pre>pair&lt;P, double&gt; mec(vector&lt;P&gt; ps) {     shuffle(all(ps), mt19937(time(0)));     P o = ps[0];     double r = 0, EPS = 1 + 1e-8;     rep(i,0,sz(ps)) if ((o - ps[i]).dist() &gt; r * EPS) {         o = ps[i], r = 0;         rep(j,0,i) if ((o - ps[j]).dist() &gt; r * EPS) {             o = (ps[i] + ps[j]) / 2;             r = (o - ps[i]).dist();             rep(k,0,j) if ((o - ps[k]).dist() &gt; r * EPS) {                 o = ccCenter(ps[i], ps[j], ps[k]);                 r = (o - ps[i]).dist();             }         }     }     return {o, r}; }</pre> |                  |

## 7.3 Polygons

| drawPolygon.py                                                                   |          |
|----------------------------------------------------------------------------------|----------|
| <p><b>Description:</b> draw single polygon</p>                                   |          |
|                                                                                  | 15 lines |
| <pre>n=int(input()) p=[tuple(map(float,input().split()))for _ in range(n)]</pre> |          |

```
p.append(p[0])
x,y=zip(*p)
a,b,c,d=min(x),max(x),min(y),max(y)
s=min(720/(b-a),720/(d-c))
f=open("polygon.svg","w")
f.write("<svg xmlns='http://www.w3.org/2000/svg' width='800'
height='800' style='background:white'>\n")
co = lambda pt: (40 + (pt[0] - a) * s, 800 - (40 + (pt[1] - c)
* s))
for i in range(n):
    x1, y1 = co(p[i])
    x2, y2 = co(p[i+1])
    f.write(f"<line x1='{x1}' y1='{y1}' x2='{x2}' y2='{y2}'
stroke='blue'/>\n")
    f.write(f"<text x='{x1}' y='{y1}' font-size='12'>{p[i][0]},
{p[i][1]}</text>\n")
f.write("</svg>")
```

InsidePolygon.h

db8d3c1, 14 lines

// check if [PQ] crosses ray from A
// definition: crossing: from y<0 to y>=0
bool crossRay(pt a, pt p, pt q) {
 return ( (q.y >= a.y) - (p.y >= a.y) ) \* orient(a,p,q) > 0;
}
// if strict, returns false when A is on the boundary
bool inPoly(vector<pt>& p, pt a, bool strict = true) {
 int nc = 0;
 for (int i = 0, n = p.size(); i < n; i++) {
 if (onSegment(p[i], p[(i+1)%n], a)) return !strict;
 nc += crossRay(a, p[i], p[(i+1)%n]);
 }
 return nc & 1; // inside if odd number of crossings
}

insideConvexPolygon.h

Description: test weather the point is in convex polygon in O(log n) time

Usage: poly a(polygon), a.in(p);

"Point.h", "onSegment.h"2cc673, 30 lines

struct poly {
 vv<pair<pt,pt>> egs; // left and right
 bool sw;
 poly(v<pt>& a): egs(2) {
 int n = sz(a);
 auto l = min\_element(all(a), [&](pt& a, pt& b) { return
 a.y < b.y; }) - begin(a);
 auto r = min\_element(all(a), [&](pt& a, pt& b) { return
 a.y > b.y; }) - begin(a);
 rep(i, l, l+n) egs[i]>=r].pb( {a[i%n], a[(i+1)%n]} );
 reverse(all(egs[1]));
 }
 bool crossRay(pt a, pt p, pt q) {
 return ( (q.y >= a.y) - (p.y >= a.y) ) \* orient(a,p,q)
 > 0;
 }
 bool in(pt a, bool strict = true) {
 int nc = 0;
 rep(i, 0, 2) {
 auto it = lower\_bound(all(egs[i]), a.y, [&](auto&
 inv, T val) {
 return max(inv.fi.y, inv.se.y) < val;
 });
 while ( it != egs[i].end() && min(it->fi.y, it->se.
 y) <= a.y ) {
 if (onSegment(it->fi, it->se, a) ) return !
 strict;

 if ( i==1 ) nc += crossRay(a, it->se, it->fi);
 else nc += crossRay(a, it->fi, it->se);
 }
 }
 }
}

PolygonArea.h

db98f1, 8 lines

// points of p should be given in counter-clockwise order
double polyArea(vector<pt>& p) {
 T area = 0;
 for (int i = 0, n = p.size(); i < n; i++) {
 area += cross(p[i], p[(i+1)%n]); // wrap back to 0 if i
 == n-1
 }
 return abs(area) / 2.0;
}

PolygonCenter.h

Description: Returns the center of mass for a polygon.

Time: O(n)

"Point.h"9706dc, 9 lines

typedef Point<double> P;
P polygonCenter(const vector<P>& v) {
 P res(0, 0); double A = 0;
 for (int i = 0, j = sz(v) - 1; i < sz(v); j = i++) {
 res = res + (v[i] + v[j]) \* v[j].cross(v[i]);
 A += v[j].cross(v[i]);
 }
 return res / A / 3;
}

PolygonCut.h

Description:

Returns a vector with the vertices of a polygon with every-thing to the left of the line going from s to e cut away.

Usage: vector<P> p = ...;
p = polygonCut(p, P(0,0), P(1,0));

"Point.h"d07181, 13 lines

typedef Point<double> P;
vector<P> polygonCut(const vector<P>& poly, P s, P e) {
 vector<P> res;
 rep(i,0,sz(poly)) {
 P cur = poly[i], prev = i ? poly[i-1] : poly.back();
 auto a = s.cross(e, cur), b = s.cross(e, prev);
 if ((a < 0) != (b < 0))
 res.push\_back(cur + (prev - cur) \* (a / (a - b)));
 if (a < 0)
 res.push\_back(cur);
 }
 return res;
}

ConvexHull.h

Description:

Returns a vector of the points of the convex hull in counter-clockwise order. Points on the edge of the hull between two other points are not considered part of the hull.

Time: O(n log n)

"Point.h"bf241e, 12 lines

v<pt> convexHull(v<pt> ps) {
 if ( sz(ps) <= 1 ) return ps;
 sort(all(ps), [&](pt& a, pt& b) { return make\_pair(a.x, a.y)
 < make\_pair(b.x, b.y); });
 v<pt> h(2\*sz(ps)+1); // sz(ps) + 1 if you don't need points
 on the boundary
 int t = 0, s = 1;

HullDiameter.h

Description: Returns the two points with max distance on a convex hull (ccw, no duplicate/collinear points).

Time: O(n)

"Point.h"c571b8, 12 lines

for (int \_ = 2; \_--; s = t--, reverse(all(ps))) for (pt p :
 ps) {
 while (t > s && orient(p, h[t-1], h[t-2]) > 0) t--;
 //>= 0 if you don't need points on the
 boundary
 h[t++] = p;
 }
 return h.resize(t), h;
}

HullDiameter.h

Description: Returns the two points with max distance on a convex hull (ccw, no duplicate/collinear points).

Time: O(n)

"Point.h"c571b8, 12 lines

typedef Point<ll> P;
array<P, 2> hullDiameter(vector<P> S) {
 int n = sz(S), j = n < 2 ? 0 : 1;
 pair<ll, array<P, 2>> res({0, {S[0], S[0]}});
 rep(i,0,j)
 for (; j = (j + 1) % n) {
 res = max(res, {{S[i] - S[j]}.dist2(), {S[i], S[j]}});
 if ((S[(j + 1) % n] - S[j]).cross(S[i + 1] - S[i]) >= 0)
 break;
 }
 return res.second;
}

PointInsideHull.h

Description: Determine whether a point t lies inside a convex hull (CCW order, with no collinear points). Returns true if point lies within the hull. If strict is true, points on the boundary aren't included.

Time: O(log N)

"Point.h", "sideOf.h", "OnSegment.h"71446b, 14 lines

typedef Point<ll> P;

bool inHull(const vector<P>& l, P p, bool strict = true) {
 int a = 1, b = sz(l) - 1, r = !strict;
 if (sz(l) < 3) return r && onSegment(l[0], l.back(), p);
 if (sideOf(l[0], l[a], l[b]) > 0) swap(a, b);
 if (sideOf(l[0], l[a], p) >= r || sideOf(l[0], l[b], p)<= -r)
 return false;
 while (abs(a - b) > 1) {
 int c = (a + b) / 2;
 (sideOf(l[0], l[c], p) > 0 ? b : a) = c;
 }
 return sgn(l[a].cross(l[b], p)) < r;
}

LineHullIntersection.h

Description: Line-convex polygon intersection. The polygon must be ccw and have no collinear points. lineHull(line, poly) returns a pair describing the intersection of a line with the polygon: • (−1, −1) if no collision, • (i, −1) if touching the corner i, • (i, i) if along side (i, i + 1), • (i, j) if crossing sides (i, i + 1) and (j, j + 1). In the last case, if a corner i is crossed, this is treated as happening on side (i, i + 1). The points are returned in the same order as the line hits the polygon. extrVertex returns the point of a hull with the max projection onto a line.

Time: O(log n)

"Point.h"7cf45b, 39 lines

#define cmp(i,j) sgn(dir.perp().cross(poly[(i)%n]-poly[(j)%n]))
#define extr(i) cmp(i + 1, i) >= 0 && cmp(i, i - 1 + n) < 0
template <class P> int extrVertex(vector<P>& poly, P dir) {
 int n = sz(poly), lo = 0, hi = n;
 if (extr(0)) return 0;
 while (lo + 1 < hi) {
 int m = (lo + hi) / 2;

```
    if (extr(m)) return m;
    int ls = cmp(lo + 1, lo), ms = cmp(m + 1, m);
    (ls < ms || (ls == ms && ls == cmp(lo, m)) ? hi : lo) = m;
}
return lo;
}

#define cmpL(i) sgn(a.cross(poly[i], b))
template <class P>
array<int, 2> lineHull(P a, P b, vector<P>& poly) {
    int endA = extrVertex(poly, (a - b).perp());
    int endB = extrVertex(poly, (b - a).perp());
    if (cmpL(endA) < 0 || cmpL(endB) > 0)
        return {-1, -1};
    array<int, 2> res;
    rep(i, 0, 2) {
        int lo = endB, hi = endA, n = sz(poly);
        while ((lo + 1) % n != hi) {
            int m = ((lo + hi + (lo < hi ? 0 : n)) / 2) % n;
            (cmpL(m) == cmpL(endB) ? lo : hi) = m;
        }
        res[i] = (lo + !cmpL(hi)) % n;
        swap(endA, endB);
    }
    if (res[0] == res[1]) return {res[0], -1};
    if (!cmpL(res[0]) && !cmpL(res[1]))
        switch ((res[0] - res[1] + sz(poly) + 1) % sz(poly)) {
            case 0: return {res[0], res[0]};
            case 2: return {res[1], res[1]};
        }
    return res;
}
```

minkowski.h  
Description: Compute Minkowski sum for two convex Polygon, return conv-  
ex in ccw

```
"Point.h", "convexHull.h"
58ccd7, 17 lines

v<pt> Minkowski(v<pt> a, v<pt> b) { // |A|,|B|>=3
    a = convexHull(a), b = convexHull(b);
    if(a.empty()) return b;
    if(b.empty()) return a;
    int n = sz(a), m = sz(b);
    v<pt> c;
    a.pb(a[0]); a.pb(a[1]);
    b.pb(b[0]); b.pb(b[1]);
    int i = 0, j = 0;
    while(i < n || j < m) {
        c.pb(a[i] + b[j]);
        T cr = cross(a[i + 1] - a[i], b[j + 1] - b[j]);
        if(cr >= 0 && i < sz(a)) i++;
        if(cr <= 0 && j < m) j++;
    }
    return c;
}
```

## 7.4 Misc. Point Set Problems

ClosestPair.h  
Description: Finds the closest pair of points.  
Time:  $\mathcal{O}(n \log n)$

```
"Point.h"
ac41a6, 17 lines

typedef Point<ll> P;
pair<P, P> closest(vector<P> v) {
    assert(sz(v) > 1);
    set<P> S;
    sort(all(v), [](P a, P b) { return a.y < b.y; });
    pair<ll, pair<P, P>> ret{LLONG_MAX, {P(), P()}};
    int j = 0;
    for (P p : v) {
```

```
P d{1 + (ll)sqrt(ret.first), 0};
while (v[j].y <= p.y - d.x) S.erase(v[j++]);
auto lo = S.lower_bound(p - d), hi = S.upper_bound(p + d);
for (; lo != hi; ++lo)
    ret = min(ret, {(lo - p).dist2(), {lo, p}});
S.insert(p);
}
return ret.second;
}
```

kdTree.h  
Description: KD-tree (2d, can be extended to 3d)

```
"Point.h"
bac5b0, 63 lines

typedef long long T;
typedef Point<T> P;
const T INF = numeric_limits<T>::max();

bool on_x(const P& a, const P& b) { return a.x < b.x; }
bool on_y(const P& a, const P& b) { return a.y < b.y; }

struct Node {
    P pt; // if this is a leaf, the single point in it
    T x0 = INF, x1 = -INF, y0 = INF, y1 = -INF; // bounds
    Node *first = 0, *second = 0;

    T distance(const P& p) { // min squared distance to a point
        T x = (p.x < x0 ? x0 : p.x > x1 ? x1 : p.x);
        T y = (p.y < y0 ? y0 : p.y > y1 ? y1 : p.y);
        return (P(x, y) - p).dist2();
    }

    Node(vector<P>&& vp) : pt(vp[0]) {
        for (P p : vp) {
            x0 = min(x0, p.x); x1 = max(x1, p.x);
            y0 = min(y0, p.y); y1 = max(y1, p.y);
        }
        if (vp.size() > 1) {
            // split on x if width >= height (not ideal...)
            sort(all(vp), x1 - x0 >= y1 - y0 ? on_x : on_y);
            // divide by taking half the array for each child (not
            // best performance with many duplicates in the middle)
            int half = sz(vp)/2;
            first = new Node({vp.begin(), vp.begin() + half});
            second = new Node({vp.begin() + half, vp.end()});
        }
    }
};
```

```
struct KDTree {
    Node* root;
    KDTree(const vector<P>& vp) : root(new Node({all(vp)})) {}

    pair<T, P> search(Node *node, const P& p) {
        if (!node->first) {
            // uncomment if we should not find the point itself:
            // if (p == node->pt) return {INF, P()};
            return make_pair((p - node->pt).dist2(), node->pt);
        }

        Node *f = node->first, *s = node->second;
        T bfirst = f->distance(p), bsec = s->distance(p);
        if (bfirst > bsec) swap(bsec, bfirst), swap(f, s);

        // search closest side first, other side if needed
        auto best = search(f, p);
        if (bsec < best.first)
            best = min(best, search(s, p));
        return best;
    }
};
```

```
// find nearest point to a point, and its squared distance
// (requires an arbitrary operator< for Point)
pair<T, P> nearest(const P& p) {
    return search(root, p);
}
};
```

FastDelaunay.h  
Description: Fast Delaunay triangulation. Each circumcircle contains none  
of the input points. There must be no duplicate points. If all points are on a  
line, no triangles will be returned. Should work for doubles as well, though  
there may be precision issues in 'circ'. Returns triangles in order {t[0][0],  
t[0][1], t[0][2], t[1][0], ...}, all counter-clockwise.  
Time:  $\mathcal{O}(n \log n)$

```
"Point.h"
eefdf5, 88 lines

typedef Point<ll> P;
typedef struct Quad* Q;
typedef __int128_t lll; // (can be ll if coords are < 2e4)
P arb(LLONG_MAX, LLONG_MAX); // not equal to any other point

struct Quad {
    Q rot, o; P p = arb; bool mark;
    P& F() { return r()->p; }
    Q& r() { return rot->rot; }
    Q prev() { return rot->o->rot; }
    Q next() { return r()->prev(); }
} *H;

bool circ(P p, P a, P b, P c) { // is p in the circumcircle?
    lll p2 = p.dist2(), A = a.dist2()-p2,
        B = b.dist2()-p2, C = c.dist2()-p2;
    return p.cross(a,b)*C + p.cross(b,c)*A + p.cross(c,a)*B > 0;
}

Q makeEdge(P orig, P dest) {
    Q r = H ? H : new Quad{new Quad{new Quad{new Quad{0}}}};
    H = r->o; r->r()->r() = r;
    rep(i, 0, 4) r = r->rot, r->p = arb, r->o = i & 1 ? r : r->r();
    r->p = orig; r->F() = dest;
    return r;
}

void splice(Q a, Q b) {
    swap(a->o->rot->o, b->o->rot->o); swap(a->o, b->o);
}

Q connect(Q a, Q b) {
    Q q = makeEdge(a->F(), b->p);
    splice(q, a->next());
    splice(q->r(), b);
    return q;
}

pair<Q, Q> rec(const vector<P>& s) {
    if (sz(s) <= 3) {
        Q a = makeEdge(s[0], s[1]), b = makeEdge(s[1], s.back());
        if (sz(s) == 2) return {a, a->r()};
        splice(a->r(), b);
        auto side = s[0].cross(s[1], s[2]);
        Q c = side ? connect(b, a) : 0;
        return {side < 0 ? c->r() : a, side < 0 ? c : b->r()};
    }

#define H(e) e->F(), e->p
#define valid(e) (e->F().cross(H(base)) > 0)
    Q A, B, ra, rb;
    int half = sz(s) / 2;
    tie(ra, A) = rec({all(s) - half});
    tie(B, rb) = rec({sz(s) - half + all(s)});
    while ((B->p.cross(H(A)) < 0 && (A = A->next())) ||
        (A->p.cross(H(B)) > 0 && (B = B->r()->o)));
}
```

```
Q base = connect(B->r(), A);
if (A->p == ra->p) ra = base->r();
if (B->p == rb->p) rb = base;

#define DEL(e, init, dir) Q e = init->dir; if (valid(e)) \
    while (circ(e->dir->F(), H(base), e->F())) { \
        Q t = e->dir; \
        splice(e, e->prev()); \
        splice(e->r(), e->r()->prev()); \
        e->o = H; H = e; e = t; \
    }
for (;;) {
    DEL(LC, base->r(), o); DEL(RC, base, prev());
    if (!valid(LC) && !valid(RC)) break;
    if (!valid(LC) || (valid(RC) && circ(H(RC), H(LC))))
        base = connect(RC, base->r());
    else
        base = connect(base->r(), LC->r());
}
return { ra, rb };
}
```

```
vector<P> triangulate(vector<P> pts) {
    sort(all(pts)); assert(unique(all(pts)) == pts.end());
    if (sz(pts) < 2) return {};
    Q e = rec(pts).first;
    vector<Q> q = {e};
    int qi = 0;
    while (e->o->F().cross(e->F(), e->p) < 0) e = e->o;
#define ADD { Q c = e; do { c->mark = 1; pts.push_back(c->p); \
    q.push_back(c->r()); c = c->next(); } while (c != e); }
    ADD; pts.clear();
    while (qi < sz(q)) if (!(e = q[qi++])->mark) ADD;
    return pts;
}
```

## 7.5 3D

### PolyhedronVolume.h

**Description:** Magic formula for the volume of a polyhedron. Faces should point outwards.

3058c3, 6 lines

```
template<class V, class L>
double signedPolyVolume(const V& p, const L& trilst) {
    double v = 0;
    for (auto i : trilst) v += p[i.a].cross(p[i.b]).dot(p[i.c]);
    return v / 6;
}
```

### Point3D.h

**Description:** Class to handle points in 3D space. T can be e.g. double or long long.

8058ae, 32 lines

```
template<class T> struct Point3D {
    typedef Point3D P;
    typedef const P& R;
    T x, y, z;
    explicit Point3D(T x=0, T y=0, T z=0) : x(x), y(y), z(z) {}
    bool operator<(R p) const {
        return tie(x, y, z) < tie(p.x, p.y, p.z); }
    bool operator==(R p) const {
        return tie(x, y, z) == tie(p.x, p.y, p.z); }
    P operator+(R p) const { return P(x+p.x, y+p.y, z+p.z); }
    P operator-(R p) const { return P(x-p.x, y-p.y, z-p.z); }
    P operator*(T d) const { return P(x*d, y*d, z*d); }
    P operator/(T d) const { return P(x/d, y/d, z/d); }
    T dot(R p) const { return x*p.x + y*p.y + z*p.z; }
    P cross(R p) const {
        return P(y*p.z - z*p.y, z*p.x - x*p.z, x*p.y - y*p.x);
    }
}
```

```
T dist2() const { return x*x + y*y + z*z; }
double dist() const { return sqrt((double)dist2()); }
//Azimuthal angle (longitude) to x-axis in interval [-pi, pi]
double phi() const { return atan2(y, x); }
//Zenith angle (latitude) to the z-axis in interval [0, pi]
double theta() const { return atan2(sqrt(x*x+y*y), z); }
P unit() const { return *this/(T)dist(); } //makes dist()==1
//returns unit vector normal to *this and p
P normal(P p) const { return cross(p).unit(); }
//returns point rotated 'angle' radians ccw around axis
P rotate(double angle, P axis) const {
    double s = sin(angle), c = cos(angle); P u = axis.unit();
    return u*dot(u)*(1-c) + (*this)*c - cross(u)*s;
}
};
```

### 3dHull.h

**Description:** Computes all faces of the 3-dimension hull of a point set. \*No four points must be coplanar\*, or else random results will be returned. All faces point outwards.

**Time:**  $\mathcal{O}(n^2)$

"Point3D.h"5b45fc, 49 lines

```
typedef Point3D<double> P3;
```

```
struct PR {
    void ins(int x) { (a == -1 ? a : b) = x; }
    void rem(int x) { (a == x ? a : b) = -1; }
    int cnt() { return (a != -1) + (b != -1); }
    int a, b;
};
```

```
struct F { P3 q; int a, b, c; };
```

```
vector<F> hull3d(const vector<P3>& A) {
    assert(sz(A) >= 4);
    vector<vector<PR>> E(sz(A), vector<PR>(sz(A), {-1, -1}));
#define E(x,y) E[f.x][f.y]
    vector<F> FS;
    auto mf = [&](int i, int j, int k, int l) {
        P3 q = (A[j] - A[i]).cross((A[k] - A[i]));
        if (q.dot(A[l]) > q.dot(A[i]))
            q = q * -1;
        F f{q, i, j, k};
        E(a,b).ins(k); E(a,c).ins(j); E(b,c).ins(i);
        FS.push_back(f);
    };
    rep(i,0,4) rep(j,i+1,4) rep(k,j+1,4)
        mf(i, j, k, 6 - i - j - k);
}
```

```
rep(i,4,sz(A)) {
    rep(j,0,sz(FS)) {
        F f = FS[j];
        if(f.q.dot(A[i]) > f.q.dot(A[f.a])) {
            E(a,b).rem(f.c);
            E(a,c).rem(f.b);
            E(b,c).rem(f.a);
            swap(FS[j--], FS.back());
            FS.pop_back();
        }
        int nw = sz(FS);
        rep(j,0,nw) {
            F f = FS[j];
#define C(a, b, c) if (E(a,b).cnt() != 2) mf(f.a, f.b, i, f.c);
            C(a, b, c); C(a, c, b); C(b, c, a);
        }
        for (F& it : FS) if ((A[it.b] - A[it.a]).cross(
            A[it.c] - A[it.a]).dot(it.q) <= 0) swap(it.c, it.b);
    }
}
```

```
return FS;
};
```

### sphericalDistance.h

**Description:** Returns the shortest distance on the sphere with radius radius between the points with azimuthal angles (longitude) f1 ( $\phi_1$ ) and f2 ( $\phi_2$ ) from x axis and zenith angles (latitude) t1 ( $\theta_1$ ) and t2 ( $\theta_2$ ) from z axis (0 = north pole). All angles measured in radians. The algorithm starts by converting the spherical coordinates to cartesian coordinates so if that is what you have you can use only the two last rows. dx\*radius is then the difference between the two points in the x direction and d\*radius is the total distance between the points.

611f07, 8 lines

```
double sphericalDistance(double f1, double t1,
    double f2, double t2, double radius) {
    double dx = sin(t2)*cos(f2) - sin(t1)*cos(f1);
    double dy = sin(t2)*sin(f2) - sin(t1)*sin(f1);
    double dz = cos(t2) - cos(t1);
    double d = sqrt(dx*dx + dy*dy + dz*dz);
    return radius*2*asin(d/2);
}
```

## Strings (8)

### KMP.h

**Description:** pi[x] computes the length of the longest prefix of s that ends at x, other than s[0...x] itself (abacaba -> 0010123). Can be used to find all occurrences of a string.

**Time:**  $\mathcal{O}(n)$

d4375c, 16 lines

```
vi pi(const string& s) {
    vi p(sz(s));
    rep(i,1,sz(s)) {
        int g = p[i-1];
        while (g && s[i] != s[g]) g = p[g-1];
        p[i] = g + (s[i] == s[g]);
    }
    return p;
}

vi match(const string& s, const string& pat) {
    vi p = pi(pat + '\0' + s), res;
    rep(i,sz(p)-sz(s),sz(p))
        if (p[i] == sz(pat)) res.push_back(i - 2 * sz(pat));
    return res;
}
```

### Zfunc.h

**Description:** z[i] computes the length of the longest common prefix of s[i:] and s, except z[0] = 0. (abacaba -> 0010301)

**Time:**  $\mathcal{O}(n)$

ee09e2, 12 lines

```
vi Z(const string& S) {
    vi z(sz(S));
    int l = -1, r = -1;
    rep(i,1,sz(S)) {
        z[i] = i >= r ? 0 : min(r - i, z[i - l]);
        while (i + z[i] < sz(S) && S[i + z[i]] == S[z[i]])
            z[i]++;
        if (i + z[i] > r)
            l = i, r = i + z[i];
    }
    return z;
}
```

Manacher.h

**Description:** For each position in a string, computes  $p[0][i]$  = half length of longest even palindrome around pos i,  $p[1][i]$  = longest odd (half rounded down).

**Time:**  $\mathcal{O}(N)$

e7ad79, 13 lines

```
array<vi, 2> manacher(const string& s) {
    int n = sz(s);
    array<vi,2> p = {vi(n+1), vi(n)};
    rep(z,0,2) for (int i=0,l=0,r=0; i < n; i++) {
        int t = r-i+!z;
        if (i<r) p[z][i] = min(t, p[z][l+t]);
        int L = i-p[z][i], R = i+p[z][i]-!z;
        while (L>=1 && R+1<n && s[L-1] == s[R+1])
            p[z][i]++, L--, R++;
        if (R>r) l=L, r=R;
    }
    return p;
}
```

MinRotation.h

**Description:** Finds the lexicographically smallest rotation of a string.

**Usage:** rotate(v.begin(), v.begin()+minRotation(v), v.end());

**Time:**  $\mathcal{O}(N)$

d07a42, 8 lines

```
int minRotation(string s) {
    int a=0, N=sz(s); s += s;
    rep(b,0,N) rep(k,0,N) {
        if (a+k == b || s[a+k] < s[b+k]) {b += max(0, k-1); break;}
        if (s[a+k] > s[b+k]) {a = b; break;}
    }
    return a;
}
```

SuffixArray.h

**Description:** Builds suffix array for a string.  $sa[i]$  is the starting index of the suffix which is  $i$ 'th in the sorted suffix array. The returned vector is of size  $n+1$ , and  $sa[0] = n$ . The lcp array contains longest common prefixes for neighbouring strings in the suffix array:  $lcp[i] = lcp(sa[i], sa[i-1])$ ,  $lcp[0] = 0$ . The input string must not contain any nul chars.

**Time:**  $\mathcal{O}(n \log n)$

635552, 22 lines

```
struct SuffixArray {
    vi sa, lcp;
    SuffixArray(string s, int lim=256) { // or vector<int>
        s.push_back(0); int n = sz(s), k = 0, a, b;
        vi x(all(s)), y(n), ws(max(n, lim));
        sa = lcp = y, iota(all(sa), 0);
        for (int j = 0, p = 0; p < n; j = max(1, j * 2), lim = p) {
            p = j, iota(all(y), n - j);
            rep(i,0,n) if (sa[i] >= j) y[p++] = sa[i] - j;
            fill(all(ws), 0);
            rep(i,0,n) ws[x[i]]++;
            rep(i,1,lim) ws[i] += ws[i - 1];
            for (int i = n; i--;) sa[--ws[x[y[i]]]] = y[i];
            swap(x, y), p = 1, x[sa[0]] = 0;
            rep(i,1,n) a = sa[i - 1], b = sa[i], x[b] =
                (y[a] == y[b] && y[a + j] == y[b + j]) ? p - 1 : p++;
        }
        for (int i = 0, j; i < n - 1; lcp[x[i++]] = k)
            for (k && k--, j = sa[x[i] - 1];
                s[i + k] == s[j + k]; k++);
    }
};
```

SAM.h

**Description:** SAM, convert string problems to DAG path problems

**Time:**  $\mathcal{O}(26N)$

3e6850, 38 lines

```
const int MAXN = 3e5+5;
```

```
int len[MAXN], suf[MAXN], cnt[MAXN], ch[26][MAXN];
int ndcnt = 0, last = 0;
//cnt(v) stores v's endpos size
int extend(int c){

    int cur = ++ndcnt;
    cnt[last] = 1;
    len[cur] = len[last] + 1;

    int p = last;
    while(p != -1 && !ch[c][p])ch[c][p] = cur,p = suf[p];

    if(p == -1)suf[cur] = 0;
    else{
        int q = ch[c][p];
        if(len[p]+1 == len[q])suf[cur] = q;
        else{
            int clone = ++ndcnt;
            len[clone] = len[p] + 1;
            suf[clone] = suf[q];
            for(int i=0;i<26;i++)ch[i][clone] = ch[i][q];

            while(p != -1 && ch[c][p] == q){
                ch[c][p] = clone;
                p = suf[p];
            }
            suf[q] = suf[cur] = clone;
        }
    }
    return last = cur;
}

int main()
{
    string s;cin>>s;
    suf[0] = -1;
    for(char c:s)extend(c-'a');
}
```

SuffixTree.h

**Description:** Ukkonen's algorithm for online suffix tree construction. Each node contains indices [l, r) into the string, and a list of child nodes. Suffixes are given by traversals of this tree, joining [l, r) substrings. The root is 0 (has l = -1, r = 0), non-existent children are -1. To get a complete tree, append a dummy symbol – otherwise it may contain an incomplete path (still useful for substring matching, though).

**Time:**  $\mathcal{O}(26N)$

aae0b8, 50 lines

```
struct SuffixTree {
    enum { N = 200010, ALPHA = 26 }; // N ~ 2*maxlen+10
    int toi(char c) { return c - 'a'; }
    string a; // v = cur node, q = cur position
    int t[N][ALPHA], l[N], r[N], p[N], s[N], v=0, q=0, m=2;

    void ukkadd(int i, int c) { suff:
        if (r[v]<=q) {
            if (t[v][c]==-1) { t[v][c]=m; l[m]=i;
                p[m++]=v; v=s[v]; q=r[v]; goto suff; }
            v=t[v][c]; q=l[v];
        }
        if (q==-1 || c==toi(a[q])) q++; else {
            l[m+1]=i; p[m+1]=m; l[m]=l[v]; r[m]=q;
            p[m]=p[v]; t[m][c]=m+1; t[m][toi(a[q])]=v;
            l[v]=q; p[v]=m; t[p[m]][toi(a[l[m]])]=m;
            v=s[p[m]]; q=l[m];
            while (q<r[m]) { v=t[v][toi(a[q])]; q+=r[v]-l[v]; }
            if (q==r[m]) s[m]=v; else s[m]=m+2;
            q=r[v]-q-r[m]; m+=2; goto suff;
        }
    }
}
```

```
SuffixTree(string a) : a(a) {
    fill(r,r+N,sz(a));
    memset(s, 0, sizeof s);
    memset(t, -1, sizeof t);
    fill(t[1],t[1]+ALPHA,0);
    s[0] = 1; l[0] = l[1] = -1; r[0] = r[1] = p[0] = p[1] = 0;
    rep(i,0,sz(a)) ukkadd(i, toi(a[i]));
}

// example: find longest common substring (uses ALPHA = 28)
pii best;
int lcs(int node, int i1, int i2, int olen) {
    if (l[node] <= i1 && i1 < r[node]) return 1;
    if (l[node] <= i2 && i2 < r[node]) return 2;
    int mask = 0, len = node ? olen + (r[node] - l[node]) : 0;
    rep(c,0,ALPHA) if (t[node][c] != -1)
        mask |= lcs(t[node][c], i1, i2, len);
    if (mask == 3)
        best = max(best, {len, r[node] - len});
    return mask;
}

static pii LCS(string s, string t) {
    SuffixTree st(s + (char)('z' + 1) + t + (char)('z' + 2));
    st.lcs(0, sz(s), sz(s) + 1 + sz(t), 0);
    return st.best;
}
};
```

AhoCorasick.h

**Description:** AC automaton, run add on all strings then run build to initialize. Replace ch with a map if alphabet size is too big

**Time:** construction takes  $\mathcal{O}(26N)$

193813, 45 lines

```
const int MAXN = 1e6, alp = 26;
int ndcnt = 0;
int fail[MAXN], dep[MAXN];
int ch[alp][MAXN], go[alp][MAXN];

vector<int> id[MAXN], udep[MAXN]; //strings node v contains
void add(string s, int x){
    int n = sz(s);
    int cur = 0;
    for(int i=0;i<n;i++){
        int c = s[i] - 'a'; //!= lab[s[i]];
        if(!ch[c][cur])ch[c][cur] = ++ndcnt;
        cur = ch[c][cur];
    }
    id[cur].push_back(x);
}

void build(){
    queue<int> q;
    for(int i=0;i<alp;i++){
        go[i][0] = ch[i][0];
        if(ch[i][0]){
            dep[ch[i][0]] = 1;
            q.push(ch[i][0]);
        }
    }
    udep[0].push_back(0);

    while(!q.empty()){
        int u = q.front();
        udep[dep[u]].push_back(u);
        q.pop();
        for(int i=0;i<alp;i++){
            int v = ch[i][u];
            if(v){
                go[i][u] = v;
            }
        }
    }
}
```

```
fail[v] = go[i][fail[u]];

dep[v] = dep[u]+1;
q.push(v);
} else {
    go[i][u] = go[i][fail[u]];
}
}
}
}
```

## Various (9)

### 9.1 Intervals

**IntervalContainer.h**  
**Description:** Add and remove intervals from a set of disjoint intervals. Will merge the added interval with any overlapping intervals in the set when adding. Intervals are [inclusive, exclusive).  
**Time:**  $\mathcal{O}(\log N)$

```
set<pii>::iterator addInterval(set<pii>& is, int L, int R) {
    if (L == R) return is.end();
    auto it = is.lower_bound({L, R}), before = it;
    while (it != is.end() && it->first <= R) {
        R = max(R, it->second);
        before = it = is.erase(it);
    }
    if (it != is.begin() && (--it)->second >= L) {
        L = min(L, it->first);
        R = max(R, it->second);
        is.erase(it);
    }
    return is.insert(before, {L,R});
}
```

```
void removeInterval(set<pii>& is, int L, int R) {
    if (L == R) return;
    auto it = addInterval(is, L, R);
    auto r2 = it->second;
    if (it->first == L) is.erase(it);
    else (int&)it->second = L;
    if (R != r2) is.emplace(R, r2);
}
```

### 9.2 Misc. algorithms

**TernarySearch.h**  
**Description:** Find the smallest  $i$  in  $[a,b]$  that maximizes  $f(i)$ , assuming that  $f(a) < \dots < f(i) \geq \dots \geq f(b)$ . To reverse which of the sides allows non-strict inequalities, change the  $<$  marked with (A) to  $\leq$ , and reverse the loop at (B). To minimize  $f$ , change it to  $>$ , also at (B).  
**Usage:** `int ind = ternSearch(0,n-1,[&](int i){return a[i]});`  
**Time:**  $\mathcal{O}(\log(b-a))$

```
template<class F>
int ternSearch(int a, int b, F f) {
    assert(a <= b);
    while (b - a >= 5) {
        int mid = (a + b) / 2;
        if (f(mid) < f(mid+1)) a = mid; // (A)
        else b = mid+1;
    }
    rep(i,a+1,b+1) if (f(a) < f(i)) a = i; // (B)
    return a;
}
```

kactlimportLIS.h

**FastKnapsack.h**  
**Description:** Given  $N$  non-negative integer weights  $w$  and a non-negative target  $t$ , computes the maximum  $S \leq t$  such that  $S$  is the sum of some subset of the weights.  
**Time:**  $\mathcal{O}(N \max(w_i))$

```
int knapsack(vi w, int t) {
    int a = 0, b = 0, x;
    while (b < sz(w) && a + w[b] <= t) a += w[b++];
    if (b == sz(w)) return a;
    int m = *max_element(all(w));
    vi u, v(2*m, -1);
    v[a+m-t] = b;
    rep(i,b,sz(w)) {
        u = v;
        rep(x,0,m) v[x+w[i]] = max(v[x+w[i]], u[x]);
        for (x = 2*m; --x > m;) rep(j, max(0,u[x]), v[x])
            v[x-w[j]] = max(v[x-w[j]], j);
    }
    for (a = t; v[a+m-t] < 0; a--);
    return a;
}
```

### 9.3 Dynamic programming

**KnuthDP.h**  
**Description:** When doing DP on intervals:  $a[i][j] = \min_{i < k < j} (a[i][k] + a[k][j]) + f(i, j)$ , where the (minimal) optimal  $k$  increases with both  $i$  and  $j$ , one can solve intervals in increasing order of length, and search  $k = p[i][j]$  for  $a[i][j]$  only between  $p[i][j-1]$  and  $p[i+1][j]$ . This is known as Knuth DP. Sufficient criteria for this are if  $f(b,c) \leq f(a,d)$  and  $f(a,c) + f(b,d) \leq f(a,d) + f(b,c)$  for all  $a \leq b \leq c \leq d$ . Consider also: LineContainer (ch. Data structures), monotone queues, ternary search.  
**Time:**  $\mathcal{O}(N^2)$

**DivideAndConquerDP.h**  
**Description:** Given  $a[i] = \min_{lo(i) \leq k < hi(i)} (f(i, k))$  where the (minimal) optimal  $k$  increases with  $i$ , computes  $a[i]$  for  $i = L..R-1$ .  
**Time:**  $\mathcal{O}((N + (hi-lo)) \log N)$

```
struct DP { // Modify at will:
    int lo(int ind) { return 0; }
    int hi(int ind) { return ind; }
    ll f(int ind, int k) { return dp[ind][k]; }
    void store(int ind, int k, ll v) { res[ind] = pii(k, v); }

    void rec(int L, int R, int LO, int HI) {
        if (L >= R) return;
        int mid = (L + R) >> 1;
        pair<ll, int> best(LLONG_MAX, LO);
        rep(k, max(LO, lo(mid)), min(HI, hi(mid)))
            best = min(best, make_pair(f(mid, k), k));
        store(mid, best.second, best.first);
        rec(L, mid, LO, best.second+1);
        rec(mid+1, R, best.second, HI);
    }
    void solve(int L, int R) { rec(L, R, INT_MIN, INT_MAX); }
};
```

### 9.4 Debugging tricks

- `signal(SIGSEGV, [](int) { _Exit(0); });`  
converts segfaults into Wrong Answers. Similarly one can catch SIGABRT (assertion failures) and SIGFPE (zero divisions). `_GLIBCXX_DEBUG` failures generate SIGABRT (or SIGSEGV on gcc 5.4.0 apparently).

- `feenableexcept(29);` kills the program on NaNs (1), 0-divs (4), infinities (8) and denormals (16).

### 9.5 Optimization tricks

`__builtin_ia32_ldmxcsr(40896);` disables denormals (which make floats 20x slower near their minimum value).

#### 9.5.1 Bit hacks

- `x & -x` is the least bit in `x`.
- `for (int x = m; x; ) { --x &= m; ... }` loops over all subset masks of `m` (except `m` itself).
- `c = x&-x, r = x+c; (((r^x) >> 2)/c) | r` is the next number after `x` with the same number of bits set.
- `rep(b,0,K) rep(i,0,(1 << K))`  
if `(i & 1 << b) D[i] += D[i^(1 << b)]`;  
computes all sums of subsets.

#### 9.5.2 Pragmas

- `#pragma GCC optimize ("Ofast")` will make GCC auto-vectorize loops and optimizes floating points better.
- `#pragma GCC target ("avx2")` can double performance of vectorized code, but causes crashes on old machines.
- `#pragma GCC optimize ("trapv")` kills the program on integer overflows (but is really slow).

**FastMod.h**  
**Description:** Compute  $a \% b$  about 5 times faster than usual, where  $b$  is constant but not known at compile time. Returns a value congruent to  $a \pmod b$  in the range  $[0, 2b)$ .

```
typedef unsigned long long ull;
struct FastMod {
    ull b, m;
    FastMod(ull b) : b(b), m(-1ULL / b) {}
    ull reduce(ull a) { // a % b + (0 or b)
        return a - (ull)((__uint128_t(m) * a) >> 64) * b;
    }
};
```

**FastInput.h**  
**Description:** Read an integer from stdin. Usage requires your program to pipe in input from file.  
**Usage:** `./a.out < input.txt`  
**Time:** About 5x as fast as `cin/scanf`.

```
inline char gc() { // like getchar()
    static char buf[1 << 16];
    static size_t bc, be;
    if (bc >= be) {
        buf[0] = 0, bc = 0;
        be = fread(buf, 1, sizeof(buf), stdin);
    }
    return buf[bc++]; // returns 0 on EOF
}
```

```
int readInt() {
    int a, c;
    while ((a = gc()) < 40);
```

```
if (a == '-') return -readInt();
while ((c = gc()) >= 48) a = a * 10 + c - 48;
return a - 48;
}
```

BumpAllocator.h

**Description:** When you need to dynamically allocate many objects and don't care about freeing them. "new X" otherwise has an overhead of something like 0.05us + 16 bytes per allocation.

```
// Either globally or in a single class:
static char buf[450 << 20];
void* operator new(size_t s) {
    static size_t i = sizeof buf;
    assert(s < i);
    return (void*)&buf[i -= s];
}
void operator delete(void*) {}
```

SmallPtr.h

**Description:** A 32-bit pointer that points into BumpAllocator memory.

```
"BumpAllocator.h"
template<class T> struct ptr {
    unsigned ind;
    ptr(T* p = 0) : ind(p ? unsigned((char*)p - buf) : 0) {
        assert(ind < sizeof buf);
    }
    T& operator*() const { return *(T*)(buf + ind); }
    T* operator->() const { return &*&this; }
    T& operator[](int a) const { return (&*&this)[a]; }
    explicit operator bool() const { return ind; }
};
```

BumpAllocatorSTL.h

**Description:** BumpAllocator for STL containers.  
**Usage:** vector<vector<int, small<int>>>> ed(N);

```
char buf[450 << 20] alignas(16);
size_t buf_ind = sizeof buf;

template<class T> struct small {
    typedef T value_type;
    small() {}
    template<class U> small(const U&) {}
    T* allocate(size_t n) {
        buf_ind -= n * sizeof(T);
        buf_ind &= 0 - alignof(T);
        return (T*)(buf + buf_ind);
    }
    void deallocate(T*, size_t) {}
};
```

SIMD.h

**Description:** Cheat sheet of SSE/AVX intrinsics, for doing arithmetic on several numbers at once. Can provide a constant factor improvement of about 4, orthogonal to loop unrolling. Operations follow the pattern "\_mm(256)?\_name\_(si(128|256)|epi(8|16|32|64)|pd|ps)". Not all are described here; grep for \_mm\_ in /usr/lib/gcc/\*/4.9/include/ for more. If AVX is unsupported, try 128-bit operations, "emmintrin.h" and #define \_\_SSE\_\_ and \_\_MMX\_\_ before including it. For aligned memory use \_mm\_malloc(size, 32) or int buf[N] alignas(32), but prefer loadu/storeu.

```
#pragma GCC target ("avx2") // or sse4.1
#include "immintrin.h"

typedef __m256i mi;
#define L(x) _mm256_loadu_si256((mi*)&(x))
```

BumpAllocator SmallPtr BumpAllocatorSTL SIMD

```
// High-level/specific methods:
// load(u)?_si256, store(u)?_si256, setzero_si256, _mm_malloc
// blendv_(epi8|ps|pd) (z?y:x), movemask_epi8 (hibits of bytes)
// i32gather_epi32(addr, x, 4): map addr[] over 32-b parts of x
// sad_epu8: sum of absolute differences of u8, outputs 4xi64
// maddubs_epi16: dot product of unsigned i7's, outputs 16xi15
// madd_epi16: dot product of signed i16's, outputs 8xi32
// extractf128_si256(, i) (256->128), cvtsi128_si32 (128->lo32)
// permute2f128_si256(x,x,1) swaps 128-bit lanes
// shuffle_epi32(x, 3*64+2*16+1*4+0) == x for each lane
// shuffle_epi8(x, y) takes a vector instead of an imm

// Methods that work with most data types (append e.g. _epi32):
// set1, blend (i8?x:y), add, adds (sat.), mullo, sub, and/or,
// andnot, abs, min, max, sign(i,x), cmp(gt|eq), unpack(lo|hi)

int sumi32(mi m) { union {int v[8]; mi m; } u; u.m = m;
    int ret = 0; rep(i,0,8) ret += u.v[i]; return ret; }
mi zero() { return _mm256_setzero_si256(); }
mi one() { return _mm256_set1_epi32(-1); }
bool all_zero(mi m) { return _mm256_testz_si256(m, m); }
bool all_one(mi m) { return _mm256_testc_si256(m, one()); }

ll example_filteredDotProduct(int n, short* a, short* b) {
    int i = 0; ll r = 0;
    mi zero = _mm256_setzero_si256(), acc = zero;
    while (i + 16 <= n) {
        mi va = L(a[i]), vb = L(b[i]); i += 16;
        va = _mm256_and_si256(_mm256_cmpgt_epi16(vb, va), va);
        mi vp = _mm256_madd_epi16(va, vb);
        acc = _mm256_add_epi64(_mm256_unpacklo_epi32(vp, zero),
                               _mm256_add_epi64(acc, _mm256_unpackhi_epi32(vp, zero)));
    }
    union {ll v[4]; mi m; } u; u.m = acc; rep(i,0,4) r += u.v[i];
    for (;i<n;++i) if (a[i] < b[i]) r += a[i]*b[i]; // <- equiv
    return r;
}
```

Mathematics (10)

10.1 Equations

$$ax^2+bx+c=0\Rightarrow x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

The extremum is given by  $x=-b/2a$ .

$$\begin{matrix}ax+by=e\\cx+dy=f\end{matrix}\Rightarrow\begin{matrix}x=\frac{ed-bf}{ad-bc}\\y=\frac{af-ec}{ad-bc}\end{matrix}$$

In general, given an equation  $Ax=b$ , the solution to a variable  $x_i$  is given by

$$x_i=\frac{\det A'_i}{\det A}$$

where  $A'_i$  is  $A$  with the  $i$ 'th column replaced by  $b$ .

10.2 Recurrences

If  $a_n=c_1a_{n-1}+\cdots+c_ka_{n-k}$ , and  $r_1,\ldots,r_k$  are distinct roots of  $x^k-c_1x^{k-1}-\cdots-c_k$ , there are  $d_1,\ldots,d_k$  s.t.

$$a_n=d_1r_1^n+\cdots+d_kr_k^n.$$

Non-distinct roots  $r$  become polynomial factors, e.g.  $a_n=(d_1n+d_2)r^n$ .

10.3 Trigonometry

$$\sin(v+w)=\sin v\cos w+\cos v\sin w$$

$$\cos(v+w)=\cos v\cos w-\sin v\sin w$$

$$\tan(v+w)=\frac{\tan v+\tan w}{1-\tan v\tan w}$$

$$\sin v+\sin w=2\sin\frac{v+w}{2}\cos\frac{v-w}{2}$$

$$\cos v+\cos w=2\cos\frac{v+w}{2}\cos\frac{v-w}{2}$$

$$(V+W)\tan(v-w)/2=(V-W)\tan(v+w)/2$$

where  $V,W$  are lengths of sides opposite angles  $v,w$ .

$$a\cos x+b\sin x=r\cos(x-\phi)$$

$$a\sin x+b\cos x=r\sin(x+\phi)$$

where  $r=\sqrt{a^2+b^2},\phi=\text{atan2}(b,a)$ .

10.4 Geometry

10.4.1 Triangles

Side lengths:  $a,b,c$

Semiperimeter:  $p=\frac{a+b+c}{2}$

Area:  $A=\sqrt{p(p-a)(p-b)(p-c)}$

Circumradius:  $R=\frac{abc}{4A}$

Inradius:  $r=\frac{A}{p}$

Length of median (divides triangle into two equal-area triangles):

$$m_a=\frac{1}{2}\sqrt{2b^2+2c^2-a^2}$$

Length of bisector (divides angles in two):

$$s_a=\sqrt{bc\left[1-\left(\frac{a}{b+c}\right)^2\right]}$$

Law of sines:  $\frac{\sin\alpha}{a}=\frac{\sin\beta}{b}=\frac{\sin\gamma}{c}=\frac{1}{2R}$

Law of cosines:  $a^2=b^2+c^2-2bc\cos\alpha$

Law of tangents:  $\frac{a+b}{a-b}=\frac{\tan\frac{\alpha+\beta}{2}}{\tan\frac{\alpha-\beta}{2}}$

10.4.2 Pick's Theorem

For a polygon with integer coordinates

$$A=i+\frac{b}{2}-1$$

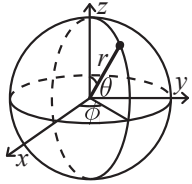
### 10.4.3 Quadrilaterals

With side lengths  $a, b, c, d$ , diagonals  $e, f$ , diagonals angle  $\theta$ , area  $A$  and magic flux  $F = b^2 + d^2 - a^2 - c^2$ :

$$4A = 2ef \cdot \sin \theta = F \tan \theta = \sqrt{4e^2 f^2 - F^2}$$

For cyclic quadrilaterals the sum of opposite angles is  $180^\circ$ ,  $ef = ac + bd$ , and  $A = \sqrt{(p-a)(p-b)(p-c)(p-d)}$ .

### 10.4.4 Spherical coordinates



$$\begin{aligned} x &= r \sin \theta \cos \phi & r &= \sqrt{x^2 + y^2 + z^2} \\ y &= r \sin \theta \sin \phi & \theta &= \arccos(z / \sqrt{x^2 + y^2 + z^2}) \\ z &= r \cos \theta & \phi &= \operatorname{atan2}(y, x) \end{aligned}$$

## 10.5 Derivatives/Integrals

$$\begin{aligned} \frac{d}{dx} \arcsin x &= \frac{1}{\sqrt{1-x^2}} & \frac{d}{dx} \arccos x &= -\frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx} \tan x &= 1 + \tan^2 x & \frac{d}{dx} \arctan x &= \frac{1}{1+x^2} \\ \int \tan ax &= -\frac{\ln |\cos ax|}{a} & \int x \sin ax &= \frac{\sin ax - ax \cos ax}{a^2} \\ \int e^{-x^2} &= \frac{\sqrt{\pi}}{2} \operatorname{erf}(x) & \int x e^{ax} dx &= \frac{e^{ax}}{a^2} (ax - 1) \end{aligned}$$

Integration by parts:

$$\int_a^b f(x)g(x)dx = [F(x)g(x)]_a^b - \int_a^b F(x)g'(x)dx$$

## 10.6 Sums

$$c^a + c^{a+1} + \dots + c^b = \frac{c^{b+1} - c^a}{c - 1}, c \neq 1$$

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(2n+1)(n+1)}{6}$$

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$$

$$1^4 + 2^4 + 3^4 + \dots + n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

## 10.7 Series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, (-\infty < x < \infty)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, (-1 < x \leq 1)$$

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{2x^3}{32} - \frac{5x^4}{128} + \dots, (-1 \leq x \leq 1)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots, (-\infty < x < \infty)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots, (-\infty < x < \infty)$$

## 10.8 Probability theory

Let  $X$  be a discrete random variable with probability  $p_X(x)$  of assuming the value  $x$ . It will then have an expected value (mean)  $\mu = \mathbb{E}(X) = \sum_x x p_X(x)$  and variance  $\sigma^2 = V(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \sum_x (x - \mathbb{E}(X))^2 p_X(x)$  where  $\sigma$  is the standard deviation. If  $X$  is instead continuous it will have a probability density function  $f_X(x)$  and the sums above will instead be integrals with  $p_X(x)$  replaced by  $f_X(x)$ .

Expectation is linear:

$$\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$$

For independent  $X$  and  $Y$ ,

$$V(aX + bY) = a^2 V(X) + b^2 V(Y).$$

### 10.8.1 Discrete distributions

#### Binomial distribution

The number of successes in  $n$  independent yes/no experiments, each which yields success with probability  $p$  is

$\operatorname{Bin}(n, p)$ ,  $n = 1, 2, \dots$ ,  $0 \leq p \leq 1$ .

$$p(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\mu = np, \sigma^2 = np(1-p)$$

$\operatorname{Bin}(n, p)$  is approximately  $\operatorname{Po}(np)$  for small  $p$ .

#### First success distribution

The number of trials needed to get the first success in independent yes/no experiments, each which yields success with probability  $p$  is  $\operatorname{Fs}(p)$ ,  $0 \leq p \leq 1$ .

$$p(k) = p(1-p)^{k-1}, k = 1, 2, \dots$$

$$\mu = \frac{1}{p}, \sigma^2 = \frac{1-p}{p^2}$$

### Poisson distribution

The number of events occurring in a fixed period of time  $t$  if these events occur with a known average rate  $\kappa$  and independently of the time since the last event is  $\operatorname{Po}(\lambda)$ ,  $\lambda = t\kappa$ .

$$p(k) = e^{-\lambda} \frac{\lambda^k}{k!}, k = 0, 1, 2, \dots$$

$$\mu = \lambda, \sigma^2 = \lambda$$

### 10.8.2 Continuous distributions

#### Uniform distribution

If the probability density function is constant between  $a$  and  $b$  and 0 elsewhere it is  $\operatorname{U}(a, b)$ ,  $a < b$ .

$$f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{otherwise} \end{cases}$$

$$\mu = \frac{a+b}{2}, \sigma^2 = \frac{(b-a)^2}{12}$$

#### Exponential distribution

The time between events in a Poisson process is  $\operatorname{Exp}(\lambda)$ ,  $\lambda > 0$ .

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$\mu = \frac{1}{\lambda}, \sigma^2 = \frac{1}{\lambda^2}$$

#### Normal distribution

Most real random values with mean  $\mu$  and variance  $\sigma^2$  are well described by  $\mathcal{N}(\mu, \sigma^2)$ ,  $\sigma > 0$ .

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

If  $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$  and  $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$  then

$$aX_1 + bX_2 + c \sim \mathcal{N}(\mu_1 + \mu_2 + c, a^2\sigma_1^2 + b^2\sigma_2^2)$$

## 10.9 Markov chains

A *Markov chain* is a discrete random process with the property that the next state depends only on the current state. Let  $X_1, X_2, \dots$  be a sequence of random variables generated by the Markov process. Then there is a transition matrix  $\mathbf{P} = (p_{ij})$ , with  $p_{ij} = \Pr(X_n = i | X_{n-1} = j)$ , and  $\mathbf{p}^{(n)} = \mathbf{P}^n \mathbf{p}^{(0)}$  is the probability distribution for  $X_n$  (i.e.,  $p_i^{(n)} = \Pr(X_n = i)$ ), where  $\mathbf{p}^{(0)}$  is the initial distribution.

$\pi$  is a stationary distribution if  $\pi = \pi \mathbf{P}$ . If the Markov chain is *irreducible* (it is possible to get to any state from any state), then  $\pi_i = \frac{1}{\mathbb{E}(T_i)}$  where  $\mathbb{E}(T_i)$  is the expected time between two visits in state  $i$ .  $\pi_j / \pi_i$  is the expected number of visits in state  $j$  between two visits in state  $i$ .

For a connected, undirected and non-bipartite graph, where the transition probability is uniform among all neighbors,  $\pi_i$  is proportional to node  $i$ 's degree.

A Markov chain is *ergodic* if the asymptotic distribution is independent of the initial distribution. A finite Markov chain is ergodic iff it is irreducible and *aperiodic* (i.e., the gcd of cycle lengths is 1).  $\lim_{k \rightarrow \infty} \mathbf{P}^k = \mathbf{1}\pi$ .

A Markov chain is an A-chain if the states can be partitioned into two sets  $\mathbf{A}$  and  $\mathbf{G}$ , such that all states in  $\mathbf{A}$  are absorbing ( $p_{ii} = 1$ ), and all states in  $\mathbf{G}$  leads to an absorbing state in  $\mathbf{A}$ . The probability for absorption in state  $i \in \mathbf{A}$ , when the initial state is  $j$ , is  $a_{ij} = p_{ij} + \sum_{k \in \mathbf{G}} a_{ik} p_{kj}$ . The expected time until absorption, when the initial state is  $i$ , is  $t_i = 1 + \sum_{k \in \mathbf{G}} p_{ki} t_k$ .

# Techniques (A)

|                                                    |           |
|----------------------------------------------------|-----------|
| techniques.txt                                     | 159 lines |
| Recursion                                          |           |
| Divide and conquer                                 |           |
| Finding interesting points in N log N              |           |
| Algorithm analysis                                 |           |
| Master theorem                                     |           |
| Amortized time complexity                          |           |
| Greedy algorithm                                   |           |
| Scheduling                                         |           |
| Max contiguous subvector sum                       |           |
| Invariants                                         |           |
| Huffman encoding                                   |           |
| Graph theory                                       |           |
| Dynamic graphs (extra book-keeping)                |           |
| Breadth first search                               |           |
| Depth first search                                 |           |
| * Normal trees / DFS trees                         |           |
| Dijkstra's algorithm                               |           |
| MST: Prim's algorithm                              |           |
| Bellman-Ford                                       |           |
| Konig's theorem and vertex cover                   |           |
| Min-cost max flow                                  |           |
| Lovasz toggle                                      |           |
| Matrix tree theorem                                |           |
| Maximal matching, general graphs                   |           |
| Hopcroft-Karp                                      |           |
| Hall's marriage theorem                            |           |
| Graphical sequences                                |           |
| Floyd-Warshall                                     |           |
| Euler cycles                                       |           |
| Flow networks                                      |           |
| * Augmenting paths                                 |           |
| * Edmonds-Karp                                     |           |
| Bipartite matching                                 |           |
| Min. path cover                                    |           |
| Topological sorting                                |           |
| Strongly connected components                      |           |
| 2-SAT                                              |           |
| Cut vertices, cut-edges and biconnected components |           |
| Edge coloring                                      |           |
| * Trees                                            |           |
| Vertex coloring                                    |           |
| * Bipartite graphs (=> trees)                      |           |
| * 3^n (special case of set cover)                  |           |
| Diameter and centroid                              |           |
| K'th shortest path                                 |           |
| Shortest cycle                                     |           |
| Dynamic programming                                |           |
| Knapsack                                           |           |
| Coin change                                        |           |
| Longest common subsequence                         |           |
| Longest increasing subsequence                     |           |
| Number of paths in a dag                           |           |
| Shortest path in a dag                             |           |
| Dynprog over intervals                             |           |
| Dynprog over subsets                               |           |
| Dynprog over probabilities                         |           |
| Dynprog over trees                                 |           |
| 3^n set cover                                      |           |
| Divide and conquer                                 |           |
| Knuth optimization                                 |           |
| Convex hull optimizations                          |           |
| RMQ (sparse table a.k.a 2^k-jumps)                 |           |
| Bitonic cycle                                      |           |
| Log partitioning (loop over most restricted)       |           |
| Combinatorics                                      |           |

|                                              |
|----------------------------------------------|
| Computation of binomial coefficients         |
| Pigeon-hole principle                        |
| Inclusion/exclusion                          |
| Catalan number                               |
| Pick's theorem                               |
| Number theory                                |
| Integer parts                                |
| Divisibility                                 |
| Euclidean algorithm                          |
| Modular arithmetic                           |
| * Modular multiplication                     |
| * Modular inverses                           |
| * Modular exponentiation by squaring         |
| Chinese remainder theorem                    |
| Fermat's little theorem                      |
| Euler's theorem                              |
| Phi function                                 |
| Frobenius number                             |
| Quadratic reciprocity                        |
| Pollard-Rho                                  |
| Miller-Rabin                                 |
| Hensel lifting                               |
| Vieta root jumping                           |
| Game theory                                  |
| Combinatorial games                          |
| Game trees                                   |
| Mini-max                                     |
| Nim                                          |
| Games on graphs                              |
| Games on graphs with loops                   |
| Grundy numbers                               |
| Bipartite games without repetition           |
| General games without repetition             |
| Alpha-beta pruning                           |
| Probability theory                           |
| Optimization                                 |
| Binary search                                |
| Ternary search                               |
| Unimodality and convex functions             |
| Binary search on derivative                  |
| Numerical methods                            |
| Numeric integration                          |
| Newton's method                              |
| Root-finding with binary/ternary search      |
| Golden section search                        |
| Matrices                                     |
| Gaussian elimination                         |
| Exponentiation by squaring                   |
| Sorting                                      |
| Radix sort                                   |
| Geometry                                     |
| Coordinates and vectors                      |
| * Cross product                              |
| * Scalar product                             |
| Convex hull                                  |
| Polygon cut                                  |
| Closest pair                                 |
| Coordinate-compression                       |
| Quadtrees                                    |
| KD-trees                                     |
| All segment-segment intersection             |
| Sweeping                                     |
| Discretization (convert to events and sweep) |
| Angle sweeping                               |
| Line sweeping                                |
| Discrete second derivatives                  |
| Strings                                      |
| Longest common substring                     |
| Palindrome subsequences                      |

|                                                       |
|-------------------------------------------------------|
| Knuth-Morris-Pratt                                    |
| Tries                                                 |
| Rolling polynomial hashes                             |
| Suffix array                                          |
| Suffix tree                                           |
| Aho-Corasick                                          |
| Manacher's algorithm                                  |
| Letter position lists                                 |
| Combinatorial search                                  |
| Meet in the middle                                    |
| Brute-force with pruning                              |
| Best-first (A*)                                       |
| Bidirectional search                                  |
| Iterative deepening DFS / A*                          |
| Data structures                                       |
| LCA (2^k-jumps in trees in general)                   |
| Pull/push-technique on trees                          |
| Heavy-light decomposition                             |
| Centroid decomposition                                |
| Lazy propagation                                      |
| Self-balancing trees                                  |
| Convex hull trick (wcipeg.com/wiki/Convex_hull_trick) |
| Monotone queues / monotone stacks / sliding queues    |
| Sliding queue using 2 stacks                          |
| Persistent segment tree                               |